Multivariate Economic Tail Risk and Scenario Analysis using the Survey of Professional Forecasters*

Manuel Schick†, Anne Opschoor‡

July 14, 2026

Abstract

This paper proposes forecasting joint tail risks for key macroeconomic indicators, GDP growth, inflation, and unemployment, using the US Survey of Professional Forecasters (SPF). By incorporating SPF consensus forecasts into the conditional mean of AR-GARCH-type models, the accuracy of univariate and multivariate predictive densities is significantly improved. Modeling a constant correlation matrix captures strong dependencies, particularly between GDP growth and unemployment. Using US data from 1990 to 2024, we show that the joint modeling framework enables scenario-based analysis in which predictive densities, conditioned on adverse developments in other variables, differ substantially from the baseline marginal distributions. The framework allows for a formal out-of-sample evaluation of joint predictive densities and a transparent assessment of conditional tail risks.

Keywords: Growth-at-Risk, Multivariate Predictive Densities, GARCH, Tail Risk, Macroeconomic Forecasting
JEL Classification: C22, C52, C53

*We would like to thank Sarah Arndt, Christian Conrad, Timo Dimitriadis, Fabian Krueger, Robinson Kruse-Becher, Sven Lehmann, Valentyn Panchenko, Julius Schoelkopf, and Bernd Schwaab, as well as seminar and conference participants at the Macro & Econometrics Seminar at Heidelberg University (December 2024), the HKMetrics Research Network Workshop at the University of Mannheim (June 2025), the Workshop on Macro- and Financial Econometrics at the School of Business, Örebro University (November 2025), the Finance@VU lunch seminar at Vrije Universiteit Amsterdam (March 2026), the 3rd CESA Workshop for Young Researchers at the FernUniversität in Hagen (April 2026), the Econometrics Department brownbag at Vrije Universiteit Amsterdam (June 2026), and the 2026 IAAE Annual Conference at Nova School of Business and Economics, Lisbon (June 2026) for their helpful comments and suggestions.

†Heidelberg University and Vrije Universiteit Amsterdam, Email: manuel.schick@awi.uni-heidelberg.de.
‡Vrije Universiteit Amsterdam and Tinbergen Institute, Email: a.opschoor@vu.nl.

1 Introduction

Policymakers routinely rely on scenario analysis to assess risks to the macroeconomy. The Federal Reserve has presented alternative scenarios to the Federal Open Market Committee in its staff briefing materials, currently the Tealbook, since the late 1960s (Herbst et al., 2026), and the International Monetary Fund regularly publishes adverse scenarios alongside the baseline projections in its *World Economic Outlook*. These multivariate exercises are not constructed as probabilistic forecasts and are therefore not subject to formal statistical evaluation.

Yet the literature on joint tail risks and multivariate scenarios remains scarce. For instance, Chavleishvili and Manganelli (2024) develop a quantile VAR model and apply it to stress-test GDP growth under adverse financial shocks.[1] Bayesian structural variants support counterfactual scenario analysis for macro-prudential policy (Chavleishvili et al., 2026). Most recently, Corradi and Llorens-Terrazas (2026) develop a parametric location-scale model to quantify joint tail risks for growth and inflation.

This paper contributes to this literature by developing a multivariate, distribution-based approach that produces scenarios as conditional predictive distributions. Our specification combines flexibility over skewness, tail thickness, and tail dependence with a parsimonious, computationally cheap structure that is straightforward to simulate from for scenario construction. We model the joint predictive density of US real GDP growth, inflation, and unemployment in a parametric AR-GARCH framework with skew-t marginals and a Student's t-copula. The conditional means incorporate consensus forecasts from the US Survey of Professional Forecasters (SPF), building on the univariate SPF-GARCH approach of Schick (2024), which we extend to a multivariate setting. Scenarios are based on conditional predictive distributions, for example the conditional quantiles of GDP growth given that inflation and unemployment both exceed their upper deciles, in the spirit of conditional Value at Risk (CoVaR; Girardi and Ergün, 2013; Adrian and Brunnermeier,

[1] The methodological foundation for conditional forecasting in vector autoregressions traces back to Waggoner and Zha (1999); see also Bańbura et al. (2015); Antolín-Díaz et al. (2021) for Bayesian implementations and extensions.

2016). Empirically, we apply the framework to US data over 1990–2024 in a real-time forecasting exercise, evaluating out-of-sample performance over the pre-pandemic period and analyzing the COVID-19 episode separately as a stress test of the framework.

Out-of-sample evaluation of the scenarios themselves would require strictly consistent scoring functions for the underlying conditional quantiles, but CoVaR does not admit such a function (Fissler and Hoga, 2024). One could in principle evaluate forecast accuracy in a specified region of interest, such as the joint tails, using censored scoring rules (Diks et al., 2011, 2014). With quarterly macroeconomic data and joint-tail conditioning events, however, very few realizations effectively contribute, raising concerns about the power of such tests in our setting. Lexicographic backtesting of VaR and CoVaR jointly (Fissler and Hoga, 2024; Dimitriadis and Hoga, 2025) faces a similar power constraint and is in its current form restricted to bivariate systems. We therefore evaluate the full joint predictive density via the logarithmic score (Gneiting and Raftery, 2007), a strictly proper rule in line with the multivariate structure of the forecast problem.

Our main findings are as follows. Incorporating SPF median forecasts significantly improves the joint predictive density of GDP growth, inflation, and unemployment across all horizons, with the largest gains emerging following the mid-quarter survey release. The data support distributional flexibility at longer horizons: GDP growth is fat-tailed but roughly symmetric, unemployment is fat-tailed and positively skewed with skewness increasing over the forecast horizon, and inflation is essentially symmetric. At the nowcast horizon, conditional on the SPF the predictive densities are close to Gaussian across indicators, consistent with the strong informational content of current-quarter survey forecasts. Conditional scenario densities differ substantially from their marginal counterparts, driven primarily by the strong negative correlation between GDP growth and unemployment, with tail dependence adding a persistent contribution after the financial crisis. For example, over the pre-pandemic sample, the conditional 10% quantile for GDP growth is 1.7–3.4 percentage points lower than the baseline across forecast horizons. The COVID-19 pandemic produced unprecedented shocks that the framework struggled to accommodate

for GDP growth and unemployment, while predictive densities for inflation remained informative through the 2021–2022 surge.

Our framework relates to several further approaches to scenario-based macroeconomic tail-risk analysis. The Growth-in-Stress framework of González-Rivera et al. (2019, 2024) constructs conditional GDP growth densities under stressed macro-financial factor scenarios via dynamic factor models. This approach focuses on GDP alone and conditions on exogenously stressed factors rather than estimating the joint distribution directly. Adrian et al. (2025) propose an entropic tilting approach that constructs full scenario densities from institutional inputs such as a baseline and a handful of reported quantiles. This provides a transparent synthesis of model-based forecasts and judgmental narratives, but still relies on externally imposed scenario paths. Bayesian VARs with stochastic volatility offer a general route to joint predictive densities and have been used to assess multivariate tail risks such as stagflation (Botelho et al., 2024), at higher computational cost than our parametric specification.

Within the univariate Growth-at-Risk (GaR) literature, downside risks to GDP growth are typically modeled by quantile regression with financial predictors, finding pronounced downside risks that respond asymmetrically to changes in financial conditions (Adrian et al., 2019; Giglio et al., 2016; Ferrara et al., 2022; Adrian et al., 2022; Delle Monache et al., 2024). The framework has been extended to other macroeconomic indicators, such as Inflation-at-Risk and Unemployment-at-Risk (Adams et al., 2021; López-Salido and Loria, 2024). Recent evidence, however, suggests that much of the time variation in these tail risks comes from movements in the conditional mean and variance rather than from time-varying skewness (Carriero et al., 2024, 2022). This motivates GARCH-based approaches, in which tail probabilities evolve directly from the estimated mean and variance without conditioning on a specific set of tail-risk predictors (Brownlees and Souza, 2021; Schick, 2024). We generalize this univariate approach to a multivariate setting for GDP growth, inflation, and unemployment.

A central ingredient of our framework is the use of consensus forecasts from the SPF,

which minimizes data requirements. Instead of assembling a large real-time predictor set from macroeconomic and financial indicators released at mixed frequencies and with varying publication lags, we incorporate the current outlook directly through SPF median forecasts. At the same time, SPF forecasts are empirically competitive with time-series models at short horizons (Stark, 2010; Faust and Wright, 2013), likely because expert aggregation quickly adapts to structural breaks and incorporates broad information sets (Ang et al., 2007). This has motivated a growing literature that uses SPF expectations for density and tail-risk forecasting, including entropic tilting with survey expectations (Krüger et al., 2017), two-step quantile regressions with SPF and financial conditions (Adams et al., 2021), and the univariate SPF-GARCH model of Schick (2024) on which we build.

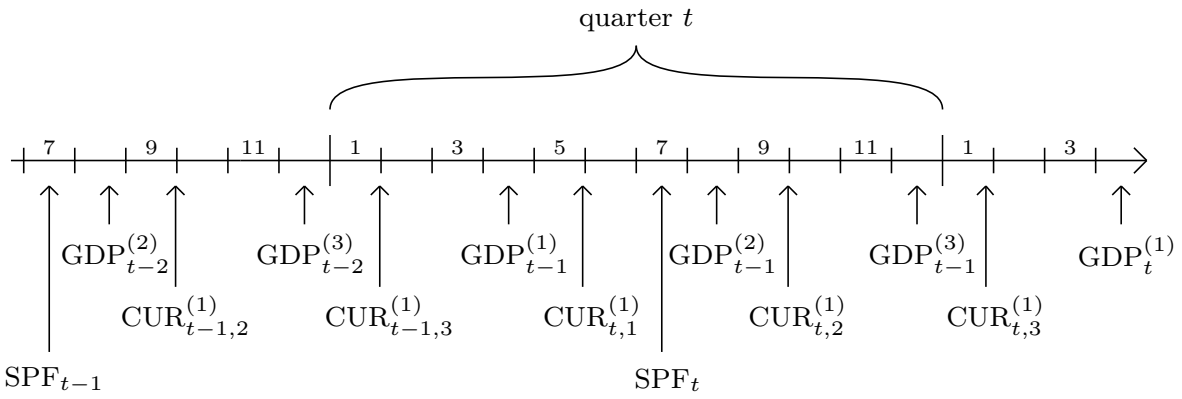
The remainder of the paper is structured as follows. Section 2 describes the real-time data and predictors. Section 3 introduces the multivariate SPF-GARCH model, its estimation, and the construction of scenarios. Section 4 outlines the forecast evaluation framework. Section 5 reports out-of-sample performance and scenario analyses over the pre-pandemic period and extends the analysis to the COVID-19 pandemic. Section 6 concludes.

2 Real-time data and predictors

To avoid look-ahead bias, we conduct a real-time out-of-sample forecasting analysis using monthly real-time vintage data on real GDP, the GDP Price Index, and the unemployment rate from the Real-Time Data Set for Macroeconomists (RTDSM) provided by the Federal Reserve Bank of Philadelphia. Information on the release dates of these monthly snapshots is available via ALFRED. Figure 1 provides an overview of the release schedule for GDP and unemployment data in weekly terms.

The Bureau of Economic Analysis (BEA) publishes the *advance* estimate of quarterly GDP at the end of the first month of the following quarter, so that the advance estimate

Figure 1: Release calendar



Notes: This graph sketches the release and revision dates of GDP and the GDP deflator, the civilian unemployment rate (CUR), and the release dates of the Survey of Professional Forecasters (SPF). GDP data refer jointly to the national accounts release of real GDP and the GDP deflator. For example, $\text{GDP}^{(1)}_{t}$ denotes the initial release for quarter t, while $\text{CUR}^{(1)}_{t,1}$ denotes the first release of the unemployment rate for the first month of quarter t. SPF releases are not subject to revisions. The numbers on the time line indicate the weeks within each quarter.

$\text{GDP}^{(1)}_{t-1}$ becomes available around week 4 of quarter t. The *preliminary* and *final* estimates, $\text{GDP}^{(2)}_{t-1}$ and $\text{GDP}^{(3)}_{t-1}$, are released one and two months later. Even the final estimate remains subject to subsequent revisions, which are reflected in the monthly vintage data. For inflation, we use the GDP Price Index, whose release coincides with that of GDP.[2]

Monthly unemployment data are released more promptly than GDP. The Civilian Unemployment Rate (CUR) is typically published on the first Friday of the following month, so that the first estimate of the unemployment rate for the first month of quarter t, $\text{CUR}^{(1)}_{t,1}$, becomes available in week 5. The RTDSM provides quarterly vintages of the monthly unemployment series, with revisions to earlier observations that are typically small and stem from seasonal adjustment updates or definition changes.

Based on these release schedules, Table 1 summarizes the latest available data vintages in each week of quarter t. In week w, we use the most recent vintage of real GDP to construct a real-time snapshot of annualized quarterly growth rates:

$$y_{s|t,w} = 400 \cdot (\log(\text{GDP}_{s|t,w}) - \log(\text{GDP}_{s-1|t,w})), \tag{1}$$

[2]Prior to 1992, the BEA reported Gross National Product (GNP) and the GNP Price Index instead of GDP.

where $\text{GDP}_{s|t,w}$ denotes quarterly GDP in quarter s, taken from the latest vintage available in week w of quarter t. The quarterly inflation rate is defined analogously,

$$\pi_{s|t,w} = 400 \cdot (\log(\text{PGDP}_{s|t,w}) - \log(\text{PGDP}_{s-1|t,w})). \tag{2}$$

For both series, the estimation sample runs over $s = 1, \ldots, t-j$, with $j = 2$ prior to the release of the previous quarter's GDP and $j = 1$ otherwise.

Table 1: Real-time vintages available in quarter t

week	Variables			
1	$\text{GDP}_{t-2\|t,1} = \text{GDP}^{(3)}_{t-2}$	$\text{CUR}_{t-1,3\|t,1} = \text{CUR}^{(1)}_{t-1,3}$	$\text{SPF}_{t+h\|t-1}$	$\text{NFCI}_{t-1,12\|t,1}$
3	$\text{GDP}_{t-2\|t,3} = \text{GDP}^{(3)}_{t-2}$	$\text{CUR}_{t-1,3\|t,3} = \text{CUR}^{(1)}_{t-1,3}$	$\text{SPF}_{t+h\|t-1}$	$\text{NFCI}_{t,2\|t,3}$
5	$\text{GDP}_{t-1\|t,5} = \text{GDP}^{(1)}_{t-1}$	$\text{CUR}_{t,1\|t,5} = \text{CUR}^{(1)}_{t,1}$	$\text{SPF}_{t+h\|t-1}$	$\text{NFCI}_{t,4\|t,5}$
7	$\text{GDP}_{t-1\|t,7} = \text{GDP}^{(1)}_{t-1}$	$\text{CUR}_{t,1\|t,7} = \text{CUR}^{(1)}_{t,1}$	$\text{SPF}_{t+h\|t}$	$\text{NFCI}_{t,6\|t,7}$
9	$\text{GDP}_{t-1\|t,9} = \text{GDP}^{(2)}_{t-1}$	$\text{CUR}_{t,2\|t,9} = \text{CUR}^{(1)}_{t,2}$	$\text{SPF}_{t+h\|t}$	$\text{NFCI}_{t,8\|t,9}$
11	$\text{GDP}_{t-1\|t,11} = \text{GDP}^{(2)}_{t-1}$	$\text{CUR}_{t,2\|t,11} = \text{CUR}^{(1)}_{t,2}$	$\text{SPF}_{t+h\|t}$	$\text{NFCI}_{t,10\|t,11}$

Notes: This table demonstrates the latest vintages of real GDP (GDP), the Civilian Unemployment Rate (CUR), h-quarters-ahead SPF projections, and NFCI data available for odd numbered weeks throughout quarter t. For instance, $\text{GDP}^{(3)}_{t-2}$ denotes the third release of GDP for quarter $t-2$ from the latest vintage in the respective week of quarter t. Similarly, $\text{CUR}^{(1)}_{t-1,3}$ denotes the first release of the CUR of the third month of quarter $t-1$ from the latest real-time vintage.

The quarterly unemployment rate is constructed as the average of the monthly Civilian Unemployment Rates,

$$u_{s|t,w} = \frac{1}{3} \sum_{m=1}^{3} \text{CUR}_{s,m|t,w}. \tag{3}$$

Because monthly figures are released in a timely manner, a first estimate $u_{t-1|t,w}$ typically becomes available about one week after the end of quarter $t-1$, so the estimation sample for unemployment generally extends to $s = t-1$. To use the within-quarter information

arriving during quarter t, we define the predictor $\tilde{u}_{s-1|t,w}$ as

$$\tilde{u}_{s-1|t,w} = \begin{cases} u_{s-1|t,w} & \text{if } w \leq 4 \\ \frac{1}{M} \sum_{m=1}^{M} \text{CUR}_{s,m|t,w} & \text{if } w > 4, \end{cases} \tag{4}$$

where $M \in \{1, 2\}$ depends on the number of monthly observations available by week w. By week $w = 5$, the unemployment rate for the first month of quarter t has typically been released ($M = 1$); once the second month's figure arrives, $\tilde{u}_{s-1|t,w}$ is the average of the first two monthly rates ($M = 2$). This predictor serves as the lag of unemployment in the mean equation (11).

Table 2 reports summary statistics for the three indicators over 1990–2024, based on the first vintage available in 2025:Q2. Including the COVID-19 pandemic in panel (b) substantially raises the volatility of all three series, reflecting the extreme downturns and recoveries of the period and the surge in inflation between 2021 and 2022.

Table 2: Summary statistics

	Panel (a): 1990 – 2019				
	Mean	Std. dev.	Min	Max	Median
Real GDP growth	2.491	2.326	-8.852	7.221	2.498
Inflation	1.974	0.910	-0.533	4.451	1.969
Unemployment rate	5.842	1.597	3.600	9.933	5.500
	Panel (b): 1990 – 2024				
	Mean	Std. dev.	Min	Max	Median
Real GDP growth	2.464	4.490	-32.957	30.156	2.691
Inflation	2.232	1.393	-1.336	8.930	2.062
Unemployment rate	5.715	1.742	3.533	13.000	5.383

Notes: Summary statistics for quarterly US real GDP growth, the GDP Price Index (inflation), and the Civilian Unemployment Rate. Panel (a) covers 1990–2019, while Panel (b) extends to 2024, capturing the COVID-19 pandemic and subsequent economic volatility. Statistics are based on the first vintage available in 2025:Q2. Real GDP growth and inflation are annualized quarter-over-quarter rates, and the unemployment rate is a quarterly average of monthly values.

Unconditional sample correlations between the three indicators (see Table A.1 in the

appendix) are modest over 1990–2019, with the strongest correlation of −0.17 between GDP growth and unemployment. On the extended sample including the COVID-19 pandemic, correlations become somewhat stronger and range from −0.28 between inflation and unemployment to 0.22 between GDP growth and inflation.

Beyond the indicators' own real-time histories, we use two predictors: the SPF and the National Financial Conditions Index (NFCI). The SPF is a quarterly survey administered since 1990 by the Federal Reserve Bank of Philadelphia, comprising approximately 40 professional forecasters (Clements et al., 2023). It is released around the middle of each quarter (typically week 6 or 7) and reports projections for the current quarter and up to four quarters ahead, including the quarter-over-quarter annualized growth rates of real GDP and the GDP Price Index and the quarterly average of the Civilian Unemployment Rate. The release schedule is illustrated in Figure 1.

We define the real-time SPF h-step-ahead forecast for indicator $i \in \{y, \pi, u\}$ in week w of quarter t as

$$\mathrm{SPF}^{i}_{t+h|t,w} = \begin{cases} \mathrm{SPF}^{i}_{t+h|t-1} & \text{if } w \leq w^{\star} \\ \mathrm{SPF}^{i}_{t+h|t} & \text{if } w > w^{\star}, \end{cases} \tag{5}$$

where $\mathrm{SPF}^{i}_{t+h|t}$ denotes the median h-step-ahead forecast from the survey released in quarter t and $w^{\star}$ the week of publication. Before the new survey is released, the most recent available forecast is from the previous quarter; for example, a one-step-ahead forecast early in quarter t uses the two-step-ahead projection $\mathrm{SPF}^{i}_{t+1|t-1}$. We also use the interquartile range (IQR) of the individual projections as a measure of forecast dispersion,

$$\mathrm{SPF}^{\mathrm{IQR},i}_{t+h|t,w} = \begin{cases} \mathrm{SPF}^{0.75,i}_{t+h|t-1} - \mathrm{SPF}^{0.25,i}_{t+h|t-1} & \text{if } w \leq w^{\star} \\ \mathrm{SPF}^{0.75,i}_{t+h|t} - \mathrm{SPF}^{0.25,i}_{t+h|t} & \text{if } w > w^{\star}, \end{cases} \tag{6}$$

where $\mathrm{SPF}^{q,i}_{t+h|t}$ denotes the q-th quantile of the cross-sectional distribution of h-step-ahead

forecasts for indicator i.[3]

The SPF also provides density forecasts in the form of histograms, which have been used to assess tail risks (Andrade et al., 2014; Ganics et al., 2024). We do not use them here because they refer to fixed events at annual frequency, whereas our framework produces fixed-horizon quarterly forecasts. Subjective density forecasts have also been shown to suffer from misspecification and poor calibration (Rossi and Sekhposyan, 2019).

The NFCI is a weekly index of US financial conditions constructed from a broad set of indicators covering money markets, debt and equity markets, and the banking system (Brave and Butters, 2012). It is standardized to mean zero and unit variance, with positive values indicating tighter-than-average conditions. The first official real-time vintage on ALFRED begins in 2011; before that date, we use the unofficial weekly real-time vintages of Amburgey and McCracken (2023), available from 1988 onward. The NFCI is released mid-week and reflects information up to the preceding Friday, implying a publication lag of approximately one week. We define the real-time quarterly NFCI in week w of quarter t as the average of the first w weekly values in each quarter s,

$$\mathrm{NFCI}_{s|t,w} = \frac{1}{w}\sum_{n=1}^{w} \mathrm{NFCI}_{s,w-n|t,w}, \tag{7}$$

where $\mathrm{NFCI}_{s,w-n|t,w}$ denotes the NFCI at week $w-n$ of quarter s, taken from the latest vintage in week w of quarter t.

3 Methodology

This section presents the methodology in three parts. Section 3.1 specifies the joint predictive density of real GDP (y), inflation (π), and unemployment (u). Section 3.2 describes the estimation, and Section 3.3 the construction of macroeconomic scenarios.

[3]The IQR is not a formal measure of forecast uncertainty (Glas, 2020), but it is a readily available proxy for cross-sectional disagreement at fixed horizons.

3.1 Forecasting models

We model the joint distribution of real GDP (y), inflation (π), and unemployment (u) by specifying flexible marginal distributions for each indicator and a copula for their dependence structure. All quantities are conditioned on the information set $\mathcal{F}_{t,w}$, comprising the real-time data and predictors available in week w of quarter t as described in Section 2. Throughout the modeling sections, we suppress the explicit vintage indexing used earlier.

For each indicator $i \in \{y, \pi, u\}$, we decompose the h-step-ahead variable $z_{t+h,i}$ as

$$z_{t+h,i} = \mu_{t+h,w,i} + \varepsilon_{t+h,w,i}, \tag{8}$$

$$\varepsilon_{t+h,w,i} = \sigma_{t+h,w,i}\, \eta_{t+h,w,i}, \tag{9}$$

where $\mu_{t+h,w,i}$ is the conditional mean, $\sigma_{t+h,w,i}$ the conditional standard deviation, and $\eta_{t+h,w,i}$ the standardized innovation with mean zero and unit variance. The standardized innovation is assumed to follow the skew-t (ST) distribution of Hansen (1994),

$$\eta_{t+h,w,i} \mid \mathcal{F}_{t,w} \sim f_{\mathrm{ST}}(0,\, 1,\, \nu^h_{w,i},\, \lambda^h_{w,i}), \tag{10}$$

where f_{ST} denotes the probability density function defined in Appendix C, $\nu^h_{w,i} > 2$ governs the tail thickness, and $\lambda^h_{w,i} \in (-1,\, 1)$ controls the skewness. We prefer Hansen's skew-t over alternative formulations, such as the one of Azzalini and Capitanio (2003), because $\mu_{t+h,w,i}$ and $\sigma^2_{t+h,w,i}$ are the conditional mean and variance of $z_{t+h,i}$.

The conditional mean and variance are modeled using an AR(1)–GARCH(1,1) framework augmented with additional predictors. Specifically, the h-step-ahead conditional mean, omitting the index i for notational convenience, is modeled as

$$\mu_{t+h,w} = c^h_w + \boldsymbol{\phi}^h_w \mathbf{x}^{\mu}_{t,w}, \tag{11}$$

The index w indicates that models are re-estimated weekly using real-time vintage data. For example, $\mu_{t+1,1}$ denotes the conditional mean for quarter $t+1$ based on information

available in week 1 of quarter t. For each indicator, $\mathbf{x}_{t,w}^{\mu}$ contains the indicator's own lag $z_{s-j,w}$. The choice of lag j aligns with direct forecasting and depends on the indicator's release schedule: if the previous quarter's macroeconomic indicator is not yet available, we set $j = 2$, otherwise $j = 1$. We also consider as predictors for the mean equation the indicator's own SPF median forecast and the NFCI.

Following Engle et al. (2013) and Conrad and Kleen (2020), we let the predictor enter the conditional variance multiplicatively,

$$\sigma_{t+h,w}^2 = g_{t+h,w}\,\psi_{t+h,w}, \tag{12}$$

$$g_{t+h,w} = (1 - \alpha_w^h - \beta_w^h) + \alpha_w^h \frac{\varepsilon_{t,w}^2}{\psi_{t,w}} + \beta_w^h\, g_{t,w}, \tag{13}$$

$$\psi_{t+h,w} = \exp(a_w^h + b_w^h x_{t,w}^{\sigma}), \tag{14}$$

where g represents the GARCH component, and ψ the volatility component associated with predictors in $x_{t,w}^{\sigma}$. The parameters satisfy the constraints $\alpha_w^h > 0$, $\beta_w^h > 0$, and $\alpha_w^h + \beta_w^h < 1$. For identification, we impose that $\omega_w^h = (1 - \alpha_w^h - \beta_w^h)$, so that the GARCH component g has unit unconditional mean and ψ captures the level of volatility. This specification ensures that the conditional variance is positive and stationary. Predictors considered for the volatility are the corresponding SPF interquartile range and the NFCI, and are standardized to have mean zero and unit variance. If equation (14) does not contain predictors in $x_{t,w}^{\sigma}$, the model simplifies to the standard GARCH(1,1) model of Bollerslev (1986).

Given the skew-t marginal distributions of each indicator, we assume the dependence across the three indicators is governed by a Student's t-copula. To this end, the standardized innovations $\eta_{t+h,w,i}$ are mapped to the unit interval via the probability integral transform

$$\tilde{\eta}_{t+h,w,i} = F_{\mathrm{ST}}(\eta_{t+h,w,i};\, \nu_{w,i}^h,\, \lambda_{w,i}^h), \tag{15}$$

where $F_{\mathrm{ST}}(\,\cdot\,;\, \nu,\, \lambda)$ denotes the cumulative distribution function (CDF) of Hansen's skew-t

distribution with mean zero and unit variance.

Defining the vector $\tilde{\boldsymbol{\eta}}_{t+h,w} = (\tilde{\eta}_{t+h,w,y},\, \tilde{\eta}_{t+h,w,\pi},\, \tilde{\eta}_{t+h,w,u})'$ of probability integral transforms, we model the dependence among y, π, and u via

$$\tilde{\boldsymbol{\eta}}_{t+h,w} \mid \mathcal{F}_{t,w} \sim \boldsymbol{C}(\,\cdot\,;\, \mathbf{R}^h_w,\, \nu^h_{w,c}), \tag{16}$$

where $\boldsymbol{C}(\,\cdot\,;\, \mathbf{R}^h_w,\, \nu^h_{w,c})$ denotes the Student's t-copula with correlation matrix $\mathbf{R}^h_w$ and $\nu^h_{w,c}$ degrees of freedom. We use a constant conditional correlation (CCC) matrix framework, i.e., the correlation matrix $\mathbf{R}^h_w$ is constant over the estimation sample but re-estimated at each forecast origin (t, w).[4] Specifically, the t-copula is defined as

$$\boldsymbol{C}(\mathbf{v};\, \mathbf{R},\, \nu) = t_3\Big(t_\nu^{-1}(v_1),\, t_\nu^{-1}(v_2),\, t_\nu^{-1}(v_3)\,;\, \mathbf{R},\, \nu\Big), \tag{17}$$

where $\mathbf{v} = (v_1, v_2, v_3)' \in (0,1)^3$, $t_3(\,\cdot\,;\, \mathbf{R},\, \nu)$ denotes the trivariate Student's t CDF with correlation $\mathbf{R}$, and t_ν^{-1} the inverse CDF of a univariate Student's t distribution, both with ν degrees of freedom.

3.2 Estimation

Estimation proceeds variable by variable, which keeps the procedure parsimonious despite the flexibility of the specification. We estimate the model separately for each horizon h, consistent with the direct forecasting approach, so that all parameters are horizon-specific.

For each indicator i, the conditional mean in equation (11) is estimated by ordinary least squares. The conditional variance in equations (12)–(14), together with the skew-t shape parameters $\nu^h_{w,i}$ and $\lambda^h_{w,i}$, is then estimated by maximum likelihood on the residuals $\varepsilon_{s,w,i}$.[5] Given the marginal fits, the standardized innovations $\eta_{t+h,w,i}$ are mapped to the probability integral transforms $\tilde{\eta}_{t+h,w,i}$ via equation (15).

[4] A dynamic conditional correlation specification (Engle, 2002) fits better in-sample but significantly worse out-of-sample, consistent with overfitting.

[5] To avoid numerical instability near the boundary $|\lambda| = 1$, we restrict $\lambda \in (-0.95, 0.95)$ in the estimation.

The dependence parameters are obtained in a second step, conditional on the marginal fits. For a given copula degrees-of-freedom parameter $\nu^h_{w,c}$, the probability integral transforms are mapped to the copula scale through the inverse Student's t CDF and standardized to unit variance,

$$\zeta_{t+h,w,i} = \sqrt{\frac{\nu^h_{w,c}-2}{\nu^h_{w,c}}}\, t^{-1}_{\nu^h_{w,c}}(\tilde{\eta}_{t+h,w,i})\,, \tag{18}$$

and $\mathbf{R}^h_w$ is set to the sample correlation matrix of $\zeta_{s,w,i}$ across indicators i, computed over the estimation sample. The matrix $\mathbf{R}^h_w$ is therefore not subject to numerical optimization but concentrated out as a function of $\nu^h_{w,c}$. The copula degrees of freedom $\nu^h_{w,c}$ is the only remaining parameter and is estimated by maximizing the copula log-likelihood.

This two-step approach ensures flexibility in the marginal distributions while keeping the joint specification tractable: the dependence structure adds a single estimated scalar, $\nu^h_{w,c}$, beyond the univariate models, which is well suited to the relatively short macroeconomic samples.

For comparison, we also estimate two nested specifications. Imposing $\lambda^h_{w,i} = 0$ for all i yields symmetric Student's t marginals, isolating fat tails from skewness. Imposing in addition $\nu^h_{w,i} \to \infty$ and $\nu^h_{w,c} \to \infty$ results in the multivariate normal distribution with Gaussian marginals and a Gaussian copula. The Gaussian case is estimated directly, replacing the Student's t transform in equation (18) and the marginal innovations with their normal counterparts. All specifications share the same two-step procedure, so that differences in fit and forecast performance reflect the distributional assumptions alone.

3.3 Predictive densities and scenario analysis

Given the joint predictive density specified in Section 3.1, we now detail the construction of baseline and scenario risk measures. Throughout, the *marginal* and *conditional* densities both refer to the predictive distribution of the macroeconomic indicators, i.e., conditional on the information set available in week w of quarter t. The marginal predictive density is

the predictive density of one indicator integrated over the others, while the conditional predictive density additionally conditions on specified events for the other indicators.

As baseline risk measures, we consider Outlook-at-Risk defined by the h-step-ahead τ-quantile of each indicator. Since we model the multivariate distribution by its marginals and a copula, the baseline quantiles are directly available from the marginal densities defined in equation (10) such that

$$Q^i_{t+h,w}(\tau) = \mu_{t+h,w,i} + \sigma_{t+h,w,i}\, F^{-1}_{ST}(\tau;\, \nu^h_{w,i},\, \lambda^h_{w,i}), \tag{19}$$

where $F^{-1}_{ST}(\,\cdot\,;\,\nu,\,\lambda)$ denotes the inverse CDF of Hansen's skew-t distribution with mean zero and unit variance.

For scenario analysis, we are primarily concerned with conditional distributions. In the spirit of CoVaR, conditional risk measures are constructed by conditioning one variable on tail events of another (Adrian and Brunnermeier, 2016). For instance, a bad scenario for GDP growth may be defined as a situation in which both inflation and unemployment are at elevated levels. Following Girardi and Ergün (2013), we condition on exceedances of the respective quantiles rather than on point values, which provides a more natural representation of severe stress events. Accordingly, bad-scenario GaR conditional on both inflation and unemployment exceeding their 90% quantiles is defined as

$$Q^y_{t+h,w}\Big(0.1 \,\Big|\, \pi \geq Q^\pi_{t+h,w}(0.9),\, u \geq Q^u_{t+h,w}(0.9)\Big)\,, \tag{20}$$

where $Q^y_{t+h,w}(\tau \,|\, \cdot)$ denotes the τ-quantile of the conditional distribution.[6] Good-scenario GaR follows analogously by reversing inequalities and considering the $(1-\tau)$-quantile, and one can similarly construct scenarios for inflation and unemployment. Conditional quantiles are computed by simulating $N = 10^6$ draws from the joint predictive density and taking the empirical τ-quantile of the subsample satisfying the conditioning event.

[6] Other risk measures are also available from our framework, such as expected shortfall and the full scenario densities. We focus on quantiles for consistency with the Growth-at-Risk literature.

4 Forecast evaluation

This section describes the construction of the 'true' realizations against which we evaluate the density forecasts, as well as the evaluation metric. An important question in forecast evaluation is which values to use as the actual outcomes. Later data vintages offer more precise estimates of the economic stance due to reduced measurement error. However, using the latest available vintage may incorporate definition changes that were unknown to the model or the SPF panelists at the time of forecasting.

To balance this trade-off, we follow Carriero et al. (2022) and define actual outcomes for quarter $t-1$ as the values available at the end of quarter t. For real GDP and the GDP Price Index, this typically includes two revisions, up to the final estimate, i.e., growth rates are computed based on the vintage containing $\text{GDP}_{t-1}^{(3)}$ in Figure 1. In rare cases where the release of $\text{GDP}_{t-1}^{(3)}$ falls into quarter $t+1$, the vintage includes only the first major revision, $\text{GDP}_{t-1}^{(2)}$.

Specifically, we define the actual outcomes for GDP growth and inflation under the assumption that each quarter consists of 12 fully observed weeks as

$$y_t = 400 \cdot (\log(\text{GDP}_{t|t+1,12}) - \log(\text{GDP}_{t-1|t+1,12})) \tag{21}$$

$$\pi_t = 400 \cdot (\log(\text{PGDP}_{t|t+1,12}) - \log(\text{PGDP}_{t-1|t+1,12})), \tag{22}$$

where $\text{GDP}_{t|t+1,12}$ and $\text{PGDP}_{t|t+1,12}$ denote real GDP and the GDP Price Index for quarter t, available in the most recent vintage at the end of quarter $t+1$.

For unemployment in quarter $t-1$, we construct actual outcomes using the quarterly vintages from the Philadelphia Fed, which capture information as available around the middle of quarter t, approximately in week 7 (see Figure 1). Thus, the actual value is defined as

$$u_t = \frac{1}{3} \sum_{m=1}^{3} \text{CUR}_{t,m|t+1,7}, \tag{23}$$

where $m = 1, 2, 3$ denotes the months within the quarter. Although minor revisions may occur, they are rare and small. As such, the precise definition of unemployment outcomes is unlikely to affect the out-of-sample evaluation.

To evaluate the forecasting models, we rely on the log-score, which is a strictly proper scoring rule (Gneiting and Raftery, 2007). The negative log-score at the realized outcomes $\mathbf{Y}_{t+h} = (y_{t+h}, \pi_{t+h}, u_{t+h})'$ is defined as

$$\text{LogS}(p^k_{t+h,w}, \mathbf{Y}_{t+h}) = -\log p^k_{t+h,w}(\mathbf{Y}_{t+h}), \tag{24}$$

where $p^k_{t+h,w}$ denotes the h-step-ahead real-time predictive density of model k at quarter t and week w. Since we report log-scores in their negative form, lower values indicate superior forecasting performance. While our primary focus is on evaluating full predictive densities using log-scores, directly assessing the conditional quantiles that define our scenarios is not feasible, as such conditional measures are not elicitable (Fissler and Hoga, 2024).

To test for differences in model performance, we use the Diebold and Mariano (1995) test. The h-step-ahead loss differential between the log-score contributions of two models k and l at forecast origin (t, w) is

$$d^{k,l}_{t+h,w} = \log p^l_{t+h,w}(\mathbf{Y}_{t+h}) - \log p^k_{t+h,w}(\mathbf{Y}_{t+h}), \tag{25}$$

where positive values indicate superior forecasting performance of model l relative to model k. We apply the Model Confidence Set (MCS) procedure (Hansen et al., 2011) to the log-score contributions to account for the multiple comparison problem, with critical values approximated via a block bootstrap with 50,000 replications and a block length of three quarters.[7]

The estimation sample is determined by the availability of the SPF, which began in 1968. For models incorporating the NFCI, the sample starts in 1973. To ensure reasonably

[7]We use the MFE Toolbox in MATLAB for the MCS procedure, see Sheppard (2009): MFE Toolbox documentation.

precise parameter estimates and allow for a sufficiently long out-of-sample period for the MCS to have statistical power, the evaluation period for the main analysis spans from 1990:Q1 to 2019:Q4, covering 120 quarters and excluding the COVID-19 pandemic in the main analysis. The pandemic period is examined separately in Section 5.3.

Due to the limited sample size, we estimate the models recursively using expanding windows. For nested models, Giacomini and White (2006) show that rolling-window estimation guarantees stationary loss differentials and a well-behaved asymptotic distribution of the MCS test statistic. Expanding-window estimation does not formally satisfy this requirement, but yields more precise parameter estimates in our relatively short macroeconomic samples. Rolling windows largely preserve the relative ranking of models but reduce the power of the MCS due to the shorter effective evaluation sample.

In addition to out-of-sample evaluation, we report pairwise likelihood-ratio tests to assess the in-sample fit of various specifications. These tests formally compare nested models, evaluating whether a more complex specification provides a statistically significant improvement over a simpler alternative.

5 Empirical results

This section presents the empirical results. We begin by evaluating the out-of-sample forecast performance over the pre-pandemic period from 1990:Q1 to 2019:Q4, which implies an initial estimation window of 85 observations and 120 out-of-sample forecasts. We report out-of-sample density forecast performance and complement these results with calibration evidence and in-sample diagnostics. We then extend the analysis to the post-2019 period and discuss the implications of the COVID-19 pandemic.

5.1 Forecast evaluation

We begin by assessing the predictive content of the SPF relative to the NFCI for the joint predictive density within an AR-GARCH framework. Table 3 presents the main results

for the evaluation period 1990:Q1 to 2019:Q4. It reports negative predictive log-scores for h-step-ahead forecasts, evaluated at the end of each respective week. Within columns, grey shading indicates inclusion in the 90% Model Confidence Set for equal predictive accuracy. For each specification, the multivariate distribution is modeled by a Student's t-copula with skew-t marginals. The first three rows report results for the benchmark AR(1)–GARCH(1,1) specification, optionally augmented with the NFCI in the conditional mean ($x^{\mu}_{t,w,i}$) or variance ($x^{\sigma}_{t,w,i}$) of indicator i. The fourth row refers to the SPF-GARCH model, which includes the SPF consensus forecast in the conditional mean. Subsequent rows build on the SPF-GARCH model, adding one predictor at a time: the NFCI in the conditional mean or variance, or the interquartile range of the SPF in the variance equation.

Table 3: Out-of-sample log-scores for AR-GARCH and SPF-GARCH specifications

	$h=3$		$h=2$		$h=1$		$h=0$	
week	4	8	4	8	4	8	4	8
AR-GARCH:	5.47	5.34	5.10	4.63	4.28	4.13	3.50	2.96
$+x^{\mu}_{t,w,i}$ = NFCI	4.69	4.64	4.60	4.53	4.09	3.97	3.32	2.93
$+x^{\sigma}_{t,w,i}$ = NFCI	5.53	5.23	4.99	4.51	4.30	4.20	3.52	3.00
SPF-GARCH:	4.90	4.46	4.56	4.23	4.00	**3.62**	3.29	2.50
$+x^{\mu}_{t,w,i}$ = NFCI	**4.64**	4.41	4.47	4.23	**3.94**	3.67	**3.22**	**2.50**
$+x^{\sigma}_{t,w,i}$ = NFCI	4.86	**4.34**	**4.37**	**4.08**	3.98	3.83	3.27	2.53
$+x^{\sigma}_{t,w,i}$ = SPF^{IQR}	4.92	4.48	4.61	4.22	4.01	3.64	3.34	2.58

Notes: This table reports average negative log-scores of the joint predictive density of GDP growth, inflation, and unemployment over the out-of-sample period 1990:Q1–2019:Q4 (120 observations). All specifications use skew-t marginals and a Student's t-copula. AR-GARCH denotes the baseline AR(1)–GARCH(1,1) model; SPF-GARCH augments the conditional mean with the SPF consensus forecast. The NFCI and the SPF interquartile range (IQR) are the real-time series constructed in Section 2. All models are estimated using an expanding window with an initial length of 85 observations. Columns indicate the forecast horizon h (in quarters) and the week w of the quarter in which forecasts are formed. Gray shading denotes inclusion in the 90% model confidence set; bold entries indicate the lowest average loss within each column.

Across all forecast horizons and weeks, the MCS consists almost exclusively of SPF-based specifications, demonstrating that the consensus forecast in the conditional mean significantly improves the multivariate predictive density. Comparing log-scores throughout the quarter reveals large improvements in the SPF-GARCH model following the release of the survey forecasts around the sixth week. Specifically, between weeks 4 and 8, forecasting

performance of the SPF-GARCH model considerably improves across forecast horizons.

Incorporating the NFCI in the conditional mean or variance of the SPF-GARCH model tends to reduce the log-scores further, especially for longer forecast horizons. However, the gains are usually small and mostly statistically insignificant, and the results do not point to a clear placement for the NFCI in either the mean or the variance equation. Adding the SPF interquartile range to the variance equation does not improve forecasting performance.

The results also demonstrate the importance of incorporating releases and revisions of macroeconomic indicators. Although the baseline AR-GARCH model is rejected from the MCS, forecast accuracy clearly improves at each horizon when moving from week 4 to week 8. Including the NFCI in the conditional mean of the AR-GARCH model further enhances predictive performance, whereas adding it to the conditional variance does not reduce predictive log-scores. The largest predictive gains appear at the longest horizon, $h = 3$, where the AR-GARCH specification with the NFCI in the conditional mean is included in the MCS and produces smaller losses than the SPF-GARCH model in week 4.

Overall, the SPF contributes the largest improvements, while the NFCI provides limited predictive gains. The SPF-GARCH model thus emerges as the preferred specification, i.e., the AR(1)–GARCH(1,1) augmented with the SPF in the conditional mean. Subsequent analyses focus on this benchmark.

To shed light on how the SPF refines the multivariate predictive densities, Table 4 decomposes the joint out-of-sample log-likelihoods of the AR-GARCH and SPF-GARCH models into the contributions of the three marginals and the copula. By construction, the Overall log-likelihood is the sum of these four components. The MCS for the Overall panel is based on the same log-scores as Table 3 and thus reproduces the comparison between AR-GARCH and SPF-GARCH.

Turning to individual series, the largest relative gains are found for unemployment, where the SPF-GARCH significantly improves the predictive density across all forecast horizons and weeks, with the single exception of $h = 1$ in week 4, where the gain is sizable

Table 4: Out-of-sample log-likelihood decomposition

	$h=3$		$h=2$		$h=1$		$h=0$	
Overall	$w=4$	$w=8$	$w=4$	$w=8$	$w=4$	$w=8$	$w=4$	$w=8$
AR-GARCH	-656.6	-640.5	-611.9	-555.8	-513.5	-495.1	-419.8	-355.6
SPF-GARCH	**-588.3**	**-535.1**	**-546.7**	**-507.7**	**-480.0**	**-434.1**	**-395.2**	**-300.5**
real GDP								
AR-GARCH	-265.7	-264.7	-262.7	-260.7	-256.3	-258.5	-248.1	-247.4
SPF-GARCH	**-265.4**	**-263.5**	**-259.9**	**-258.3**	**-255.3**	**-252.9**	**-242.6**	**-234.1**
Inflation								
AR-GARCH	-186.9	-193.0	-203.6	-201.7	-192.7	-195.9	-179.0	-181.3
SPF-GARCH	**-185.2**	**-184.8**	**-201.4**	**-195.7**	**-177.8**	**-176.4**	**-166.1**	**-153.8**
Unemployment								
AR-GARCH	-194.6	-178.6	-135.0	-89.8	-56.7	-38.8	6.5	68.7
SPF-GARCH	**-145.3**	**-99.2**	**-95.9**	**-63.1**	**-50.6**	**-9.5**	**12.4**	**85.1**
t copula								
AR-GARCH	-9.42	-4.20	-10.58	-3.65	-7.86	-1.99	0.76	**4.34**
SPF-GARCH	**7.69**	**12.36**	**10.50**	**9.39**	**3.69**	**4.83**	**1.12**	2.37

Notes: This table decomposes the out-of-sample log-likelihood of the joint predictive density (Overall) into the contributions of the skew-t marginals for GDP growth, inflation, and unemployment, and the Student's t-copula. By construction, the Overall log-likelihood equals the sum of the four panels below. AR-GARCH denotes the baseline AR(1)–GARCH(1,1) model; SPF-GARCH augments the conditional mean with the SPF consensus forecast. The evaluation period is 1990:Q1–2019:Q4 (120 observations), and all models are estimated using an expanding window with an initial length of 85 observations. Columns indicate the forecast horizon h (in quarters) and the week w of the quarter in which forecasts are formed. Higher log-likelihoods indicate better predictive performance. The Model Confidence Set is computed separately within each panel; gray shading denotes inclusion in the 90% set, and bold entries indicate the highest log-likelihood within each column.

but not statistically significant.

For inflation, the picture is more nuanced. Improvements are consistently significant at the nowcast and one-step-ahead horizons. The mid-quarter release of the SPF enhances predictive performance also at longer horizons in week 8, although at $h = 2$ the gain falls short of significance. Early in the quarter, however, the SPF contributes little at the two- and three-step-ahead forecast horizons. Before the mid-quarter survey release, the SPF-GARCH model relies on the survey projections from the previous quarter, while the AR-GARCH model already incorporates the new GDP Price Index release. As a result, the differences are small and not statistically significant at $h = 3$ and $h = 2$ in week 4, suggesting limits to the SPF's predictive contribution for inflation at longer horizons before

the mid-quarter release.

For GDP growth, the SPF-GARCH model significantly improves the nowcast and one-step-ahead predictive densities in week 8, after the mid-quarter release of the SPF. In week 4, the AR-GARCH model benefits from the release of the advance GDP estimate, which was not yet available to the SPF respondents in the previous quarter, so the SPF-GARCH gain is sizable but not statistically significant.

At longer horizons, the picture is more striking. The SPF-GARCH consistently produces smaller losses than the AR-GARCH at $h = 2$ and $h = 3$, but the differences are small and never reach statistical significance, regardless of the week of the quarter. This points to limits to the predictability of GDP growth at the two- and three-step-ahead horizons, in line with Stark (2010), who documents that the SPF's predictive accuracy for GDP declines quickly with horizon.

Lastly, the bottom panel of Table 4 reports the copula contribution to the joint log-likelihood, measuring how much the Student's t-copula adds relative to assuming independence across indicators. Under the AR-GARCH model, the copula contribution is negative at $h = 1$ and beyond, indicating that the inferred dependence structure does not generalize out-of-sample. Under the SPF-GARCH model, by contrast, the copula contribution is large and positive at $h = 1$ and beyond, and only the SPF-GARCH is included in the MCS for the copula panel. At the nowcast horizon, copula contributions are small and similar across models.

The out-of-sample log-likelihoods measure the overall predictive accuracy. We complement the analysis by assessing the calibration of the marginal distributions. The predictive densities of the SPF-GARCH model are well calibrated for all indicators and across horizons, as shown by the probability integral transforms in Figure B.1 in the appendix. Combined with the favorable out-of-sample log-scores, this confirms that the SPF-GARCH model produces reliable density forecasts for all three indicators.

Next, we examine the role of distributional assumptions. Table 5 compares the AR-GARCH and SPF-GARCH models across three distributional specifications: a mul-

tivariate normal (Gaussian copula with normal marginals), a symmetric Student's t specification, and the main specification with skew-t marginals and a Student's t-copula. The AR-GARCH model is rejected from the MCS in every column, confirming that the predictive gains from the SPF consensus forecast hold across distributional assumptions. Within the SPF-GARCH model, the three distributional specifications yield similar and statistically indistinguishable out-of-sample losses at horizons up to two quarters ahead. At $h = 3$, however, both Student's t-copula specifications significantly outperform the multivariate normal.

Table 5: Out-of-sample log-scores for alternative distributional specifications

			$h=3$		$h=2$		$h=1$		$h=0$	
			Week of quarter							
Model	Copula	Marginal	4	8	4	8	4	8	4	8
AR-GARCH	Normal	Normal	5.92	5.86	5.30	5.04	4.48	4.32	3.65	3.09
	t	t	5.77	5.71	5.06	4.80	4.44	4.23	3.56	2.97
	t	Skew-t	5.47	5.34	5.10	4.63	4.28	4.13	3.50	2.96
SPF-GARCH	Normal	Normal	5.60	5.07	5.00	4.39	4.17	3.75	3.38	2.55
	t	t	5.37	4.62	4.67	**4.18**	4.04	3.63	3.33	2.52
	t	Skew-t	**4.90**	**4.46**	**4.56**	4.23	**4.00**	**3.62**	**3.29**	**2.50**

Notes: This table reports average negative log-scores of the joint predictive density of GDP growth, inflation, and unemployment over the out-of-sample period 1990:Q1–2019:Q4 (120 observations). AR-GARCH denotes the baseline AR(1)–GARCH(1,1) model; SPF-GARCH augments the conditional mean with the SPF consensus forecast. The three distributional specifications are nested: a Gaussian copula with normal marginals (Normal/Normal), a Student's t-copula with symmetric Student's t marginals (t/t), and a Student's t-copula with skew-t marginals (t/Skew-t). All models are estimated using an expanding window with an initial length of 85 observations. Columns are grouped by the forecast horizon h (in quarters). Gray shading denotes inclusion in the 90% model confidence set; bold entries indicate the lowest average loss within each column.

In-sample, the evidence for distributional flexibility is much stronger. Table A.2 in the appendix reports in-sample log-likelihoods of the SPF-GARCH model under the three nested specifications. Across all forecast horizons, the Student's t-copula with skew-t marginals significantly improves model fit relative to the multivariate normal baseline, with the gains increasing in the forecast horizon. The skew-t marginals are clearly preferred from an in-sample perspective while imposing no out-of-sample costs. The copula approach with skewness in the marginals thus provides a good trade-off between flexibility and

parsimony.

Figures B.2 and B.3 plot the degrees-of-freedom and skewness parameters of the SPF-GARCH model from the expanding-window scheme, evaluated in week 8 of each quarter. The estimates are noisy and respond strongly to crisis periods, but several patterns are robust.

The copula degrees-of-freedom parameter shows weak tail dependence in the mid-2000s, but drops sharply around the financial crisis and remains low throughout the post-crisis period, indicating persistent tail dependence across the three indicators. For GDP growth, the marginal degrees-of-freedom parameter is consistently below 20 for the one-step-ahead horizon and beyond; only at the nowcast horizon does GDP appear close to normality. Unemployment becomes similarly fat-tailed across most horizons from 2000 onward, except for $h = 2$, where the degrees-of-freedom parameter remains moderate until 2010. Inflation is generally not fat-tailed, especially at longer forecast horizons.

In addition to fat tails, unemployment exhibits positive skewness which is increasing with the forecast horizon, consistent with the asymmetric risk of unemployment spikes and the lower bound on the unemployment rate. In contrast, GDP growth is essentially symmetric throughout the sample at the nowcast and one-step-ahead horizons, and only mildly skewed at $h = 2$ and $h = 3$. This pattern is consistent with prior evidence that downside risks to GDP growth are driven by fat-tailed disturbances and time-varying conditional variance rather than by skewness (Carriero et al., 2024, 2022). Inflation is positively skewed at shorter horizons (with λ approaching 0.5 in some periods) but close to symmetric at $h = 3$.

Taken together, the SPF consensus forecast is the key predictor for the conditional mean of each indicator, refining both the marginal and joint predictive densities. Modeling the multivariate distribution by a tail-dependent Student's t-copula with flexible skew-t marginals adds value in-sample without compromising out-of-sample accuracy. We therefore use this model in the scenario analysis and the COVID-19 period that follow.

5.2 Scenario-based risk assessment

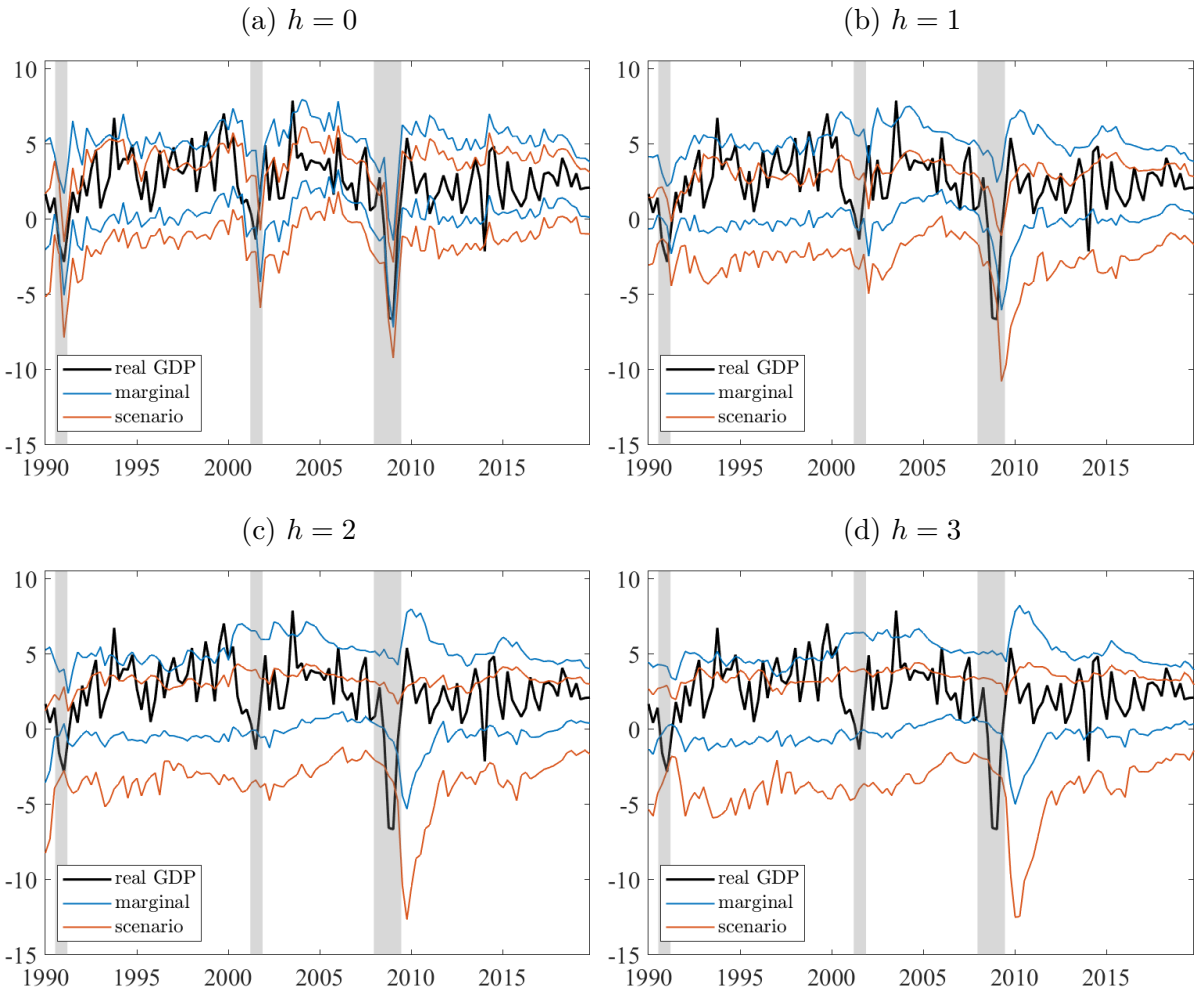
The SPF-GARCH model with skew-t marginals and a Student's t-copula, established as the preferred specification in the previous section, provides the joint predictive density required for scenario analysis. We exploit this joint structure to construct conditional risk measures, in which the predictive density of one indicator is evaluated under adverse outcomes in the others.

We begin with real GDP growth and define a downside scenario as the conditional distribution of y_{t+h} given that both inflation and unemployment exceed their 90% quantiles. This supply-shock-like scenario captures the impact of simultaneous adverse surprises in prices and the labor market. Figure 2 reports the 10% and 90% quantiles of the marginal predictive density (blue) and of the conditional density under the adverse scenario (red) over the evaluation period 1990:Q1–2019:Q4, evaluated at week 8 within each quarter (after the SPF mid-quarter release).

Across all horizons, the scenario distributions are shifted downward relative to the marginal predictive density. For the nowcast, the conditional 10% quantile lies on average 1.7 percentage points below the marginal, and this gap widens with the forecast horizon up to 3.4 percentage points at $h = 3$. At shorter horizons, the predictive densities track realized GDP growth closely, while the two- and three-step-ahead densities are wider and smoother. In particular, at the nowcast horizon, the scenario GaR captured the severity of the global financial crisis recession, although the marginal 10% quantile also signaled elevated downside risk. After the recession, both the conditional and marginal nowcast densities indicated a quick recovery following two quarters of large negative growth. By contrast, at $h \geq 1$, the gap between the marginal and scenario GaR widens substantially around the financial crisis, reflecting larger forecast errors at longer horizons.

Figure 3 helps explain these shifts. The dominant relationship is the strong negative conditional correlation between the standardized residuals of GDP growth and unemployment, ranging from around -0.3 at the nowcast to roughly -0.5 at $h = 1$ and $h = 2$, and around -0.4 at $h = 3$. This negative correlation is the main driver of the downward

Figure 2: Downside GDP scenarios based on the SPF-GARCH model

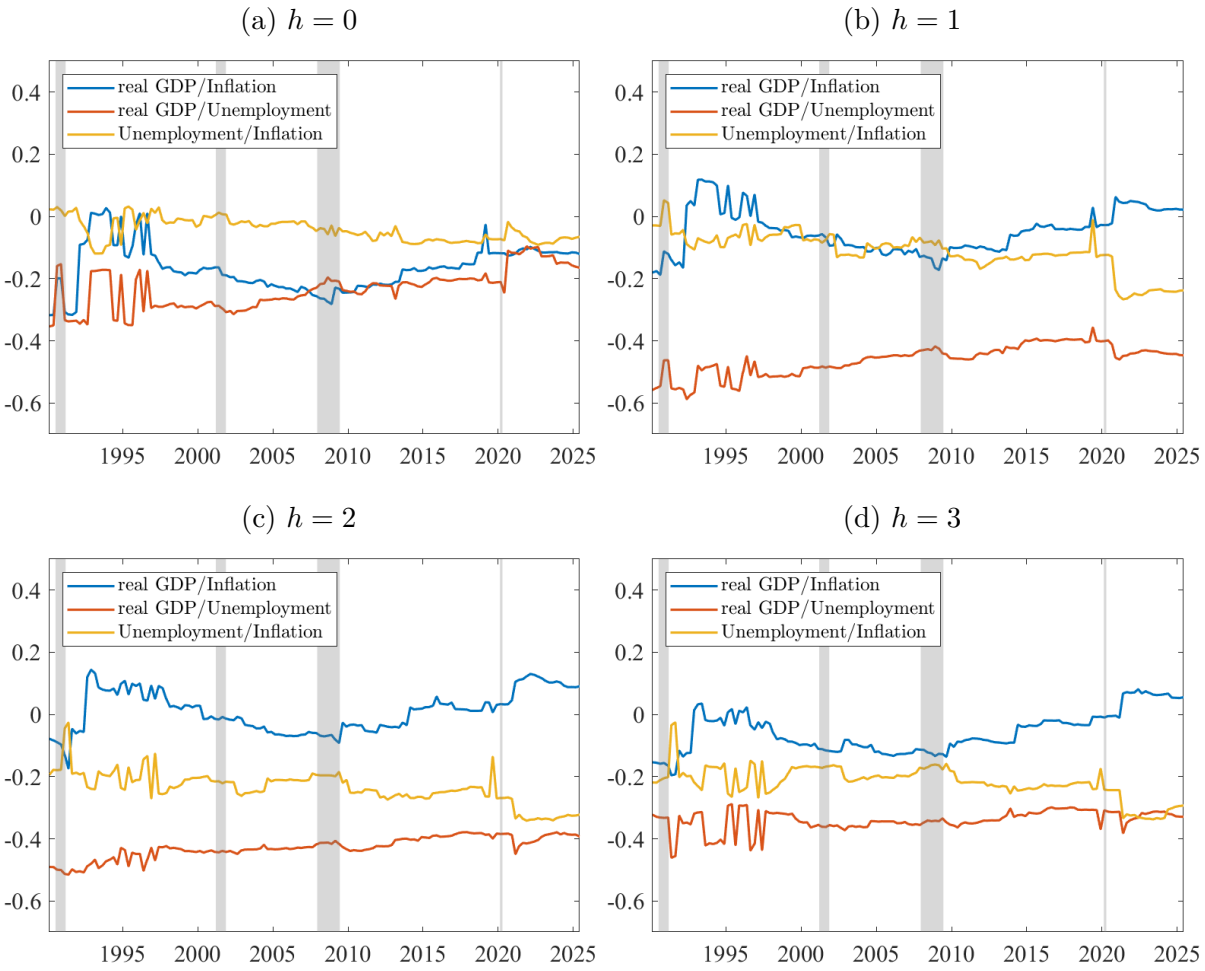


Notes: This figure plots the 10th and 90th percentiles of the marginal predictive density for real GDP growth (blue lines) and the corresponding conditional percentiles when both inflation and unemployment are in their upper deciles (red lines), i.e., $Q^{y}_{t+h,w}(\tau \,|\, \pi \geq Q^{\pi}_{t+h,w}(0.9),\, u \geq Q^{u}_{t+h,w}(0.9))$ for $\tau \in \{0.1, 0.9\}$. The black line depicts realized quarterly annualized GDP growth. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. All forecasts are from the SPF-GARCH model with skew-t marginals and a Student's t-copula. The sample period is 1990:Q1–2019:Q4, and the model is estimated using an expanding window with an initial length of 85 observations. Vertical gray bands indicate NBER recessions.

shift in GDP under the adverse scenario, and the strengthening of the correlation with horizon explains why the scenario gap widens from 1.7 to 3.4 percentage points between $h = 0$ and $h = 3$. The correlation between GDP growth and inflation is close to zero at longer horizons, contributing little to the joint scenario. Overall, the expanding-window correlation estimates are relatively constant over time.

We now turn to the role of the SPF in the conditional mean. Figure 4 shows the

Figure 3: Estimated correlations from the SPF-GARCH model

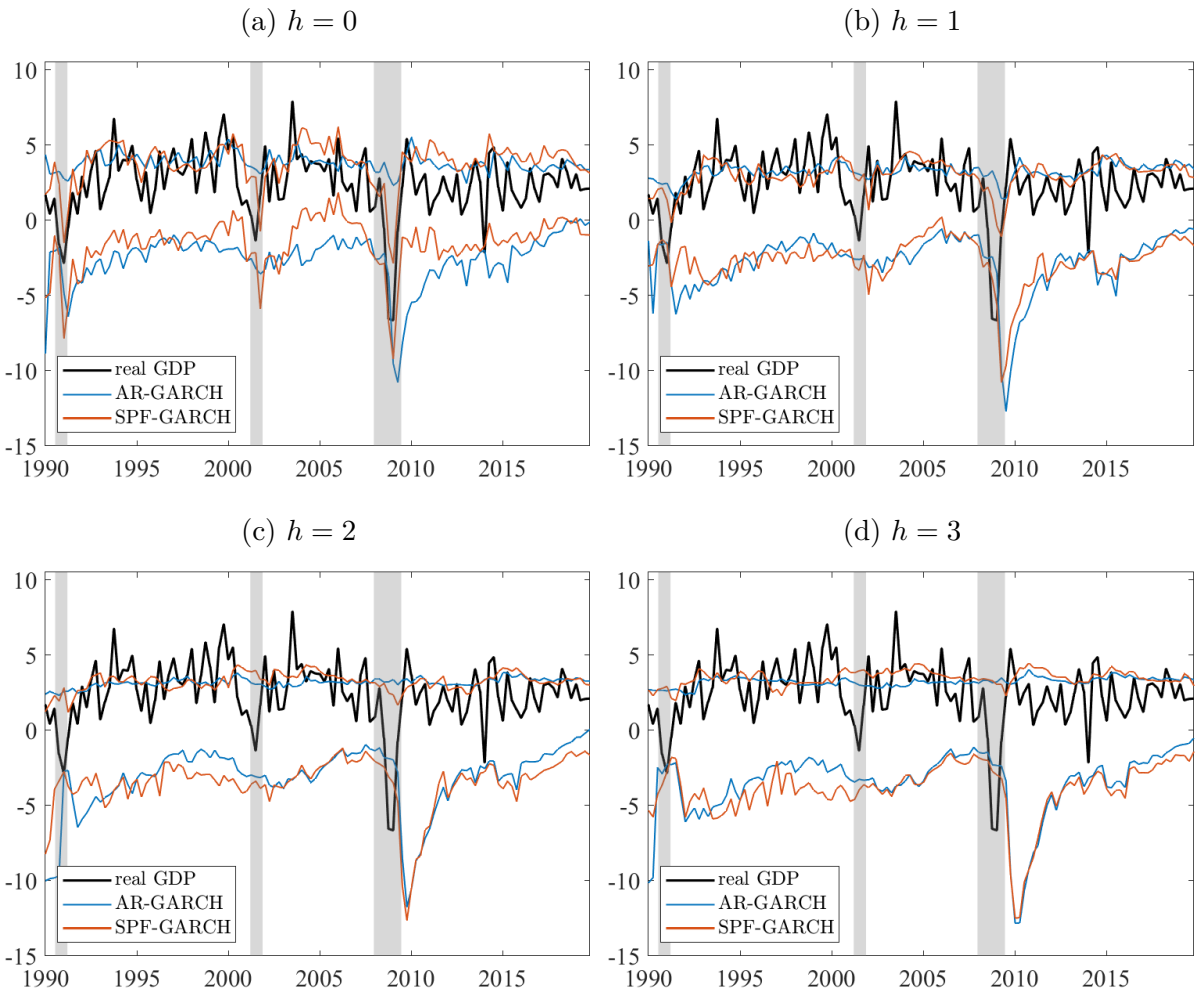


Notes: This figure plots the constant conditional correlation (CCC) coefficients for GDP growth, inflation, and unemployment estimated from the SPF-GARCH-CCC model estimated using an expanding window. Panel (a) shows results for the one-step-ahead distribution ($h = 1$) and panel (b) for the nowcast distribution ($h = 0$), each evaluated at week $w = 8$ within the quarter. The sample period is 1990:Q1–2025:Q1. Vertical gray bands indicate NBER recessions.

difference in scenario densities between the AR-GARCH and the SPF-GARCH models. Compared to the benchmark AR-GARCH specification, incorporating the current macroeconomic outlook via the SPF consensus forecast clearly shifts the conditional distribution at the nowcast and one-step-ahead horizons. For instance, following the global financial crisis, both models indicate substantial downside risk, but the SPF-based density adjusts more quickly and appears less pessimistic, reflecting a faster recovery in expectations. The largest differences appear at the nowcast horizon, where the SPF delivers the greatest gains in the conditional mean of GDP growth, shifting the quantiles of the entire predictive

density.

Figure 4: Downside GDP scenarios under AR-GARCH and SPF-GARCH models



Notes: This figure plots the 10th and 90th percentiles of the conditional predictive density for real GDP growth when both inflation and unemployment are in their upper deciles, i.e., $Q^{y}_{t+h,w}(\tau \,|\, \pi \geq Q^{\pi}_{t+h,w}(0.9),\, u \geq Q^{u}_{t+h,w}(0.9))$ for $\tau \in \{0.1, 0.9\}$, from the AR-GARCH model (blue lines) and the SPF-GARCH model (red lines). The black line depicts realized quarterly annualized GDP growth. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. Both models use skew-t marginals and a Student's t-copula. The sample period is 1990:Q1–2019:Q4, and the models are estimated using an expanding window with an initial length of 85 observations. Vertical gray bands indicate NBER recessions.

At longer horizons, these differences fade. In particular, after the global financial crisis shock, both models produce downside GaR estimates that are virtually identical. The SPF-GARCH framework can still be used for scenario analysis at horizons $h \geq 2$, but the resulting scenario distributions for GDP are similar to those from the AR-GARCH model, consistent with the limited predictive gains of the consensus forecast at these horizons

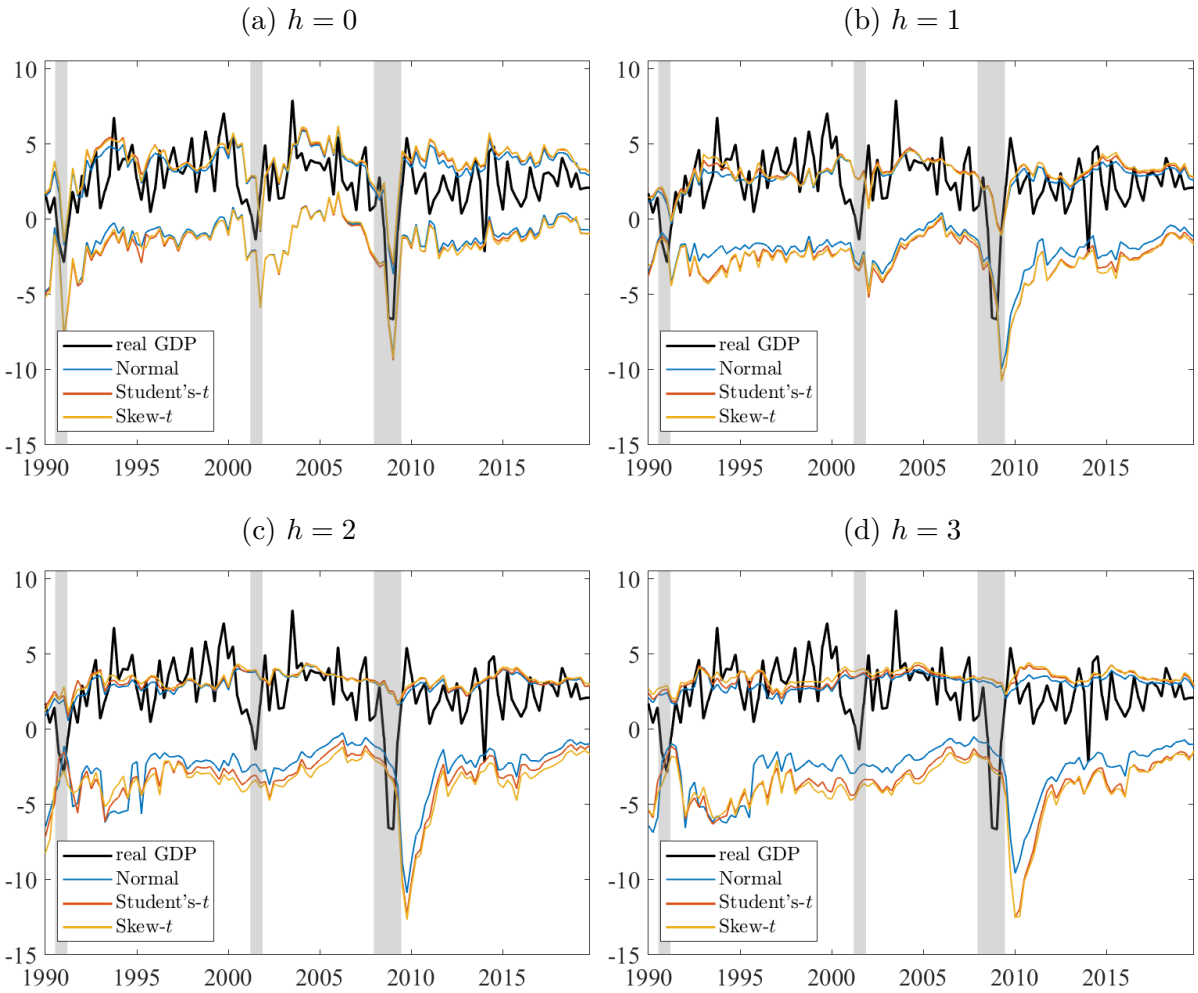
documented in the previous section.

We next examine how the distributional specification affects the scenario densities. Figure 5 assesses the relevance of various distributional assumptions. Each panel compares scenario densities using a multivariate normal distribution (Normal), a Student's t-copula with symmetric t marginals (Student's t), and a Student's t-copula with skew-t marginals (Skew-t). Two effects shape the differences across specifications. The first is fat tails, which lower the conditional 10% quantile relative to the normal benchmark by about 0.2 percentage points at the nowcast, 0.6 at $h = 1$ and $h = 2$, and 0.9 at $h = 3$. Thus, tail thickness increases with the forecast horizon. The second effect is skewness, which contributes essentially nothing at $h = 0$ and $h = 1$ but adds a further 0.2 to 0.3 percentage points downward at $h = 2$ and $h = 3$. This pattern is consistent with the earlier finding that GDP growth is approximately symmetric at short horizons and only mildly skewed at longer horizons.

Next, Figure B.4 presents upside scenarios for inflation. Motivated by the signs of the correlations among standardized residuals in Figure 3, we define an upside scenario as the conditional distribution of π_{t+h} given that both GDP growth and unemployment are in their lower deciles. As for GDP, this conditioning isolates a supply-shock-like setting of joint surprises—weak growth combined with tight labor market conditions—which the copula maps to an upward shift in the inflation distribution. The mechanism shifts with horizon: at the nowcast, the negative correlation between inflation and GDP growth does most of the work, while at $h \geq 1$ the negative correlation between inflation and unemployment becomes the dominant Phillips-curve-like channel. The scenario 90% quantile is on average shifted upward by 0.7 percentage points at $h = 0$, rising to 1.2 percentage points at $h = 3$.

Figure B.5 shows that the SPF-GARCH produces tighter scenario densities than the AR-GARCH at short horizons, although the difference becomes small at $h = 3$. Figure B.6 compares scenario densities across distributional specifications. While the three specifications produce visually similar scenario densities, fat-tails shift the conditional 90% quantile upward relative to the normal benchmark by 0.2 to 0.4 percentage points across

Figure 5: Downside GDP scenarios across distributional specifications



Notes: This figure plots the 10th and 90th percentiles of the conditional predictive density for real GDP growth when both inflation and unemployment are in their upper deciles, i.e., $Q^{y}_{t+h,w}(\tau \,|\, \pi \geq Q^{\pi}_{t+h,w}(0.9),\, u \geq Q^{u}_{t+h,w}(0.9))$ for $\tau \in \{0.1, 0.9\}$, across three distributional specifications of the SPF-GARCH model: a Gaussian copula with normal marginals (blue), a Student's t-copula with symmetric Student's t marginals (red), and a Student's t-copula with skew-t marginals (yellow). The black line depicts realized quarterly annualized GDP growth. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. The sample period is 1990:Q1–2019:Q4, and the models are estimated using an expanding window with an initial length of 85 observations. Vertical gray bands indicate NBER recessions.

horizons, with skewness contributing an additional 0.1 to 0.3 percentage points.

Finally, Figure B.7 shows scenario densities for unemployment. We define an upside scenario for unemployment as a demand-shock-like setting with both GDP growth and inflation being below their 10% quantiles. At the nowcast horizon, the predictive density of unemployment is extremely tight, indicating small forecast errors and leaving little room for the scenario to shift the conditional distribution. The 90% quantile is shifted

upward on average by only 0.1 percentage points. At $h \geq 1$, however, the conditional 90% quantile differs from the baseline on average by 0.5 to 1.2 percentage points, with the size of the shift increasing in the forecast horizon. This pattern is driven primarily by the strong negative correlation between unemployment and GDP growth, which is the dominant cross-correlation across all horizons.

In the previous section, the SPF improved the predictive density for unemployment across all forecast horizons. Consequently, in Figure B.8, the conditional densities from the SPF-GARCH model are tighter than those from the AR-GARCH throughout. Figure B.9 highlights the prominent role of skewness for the scenario densities. At the nowcast horizon, the three specifications are visually indistinguishable. At $h \geq 1$, however, the 90% quantile of the skew-t specification shifts upward relative to the normal by 0.2 to 0.7 percentage points across horizons. Fat tails play a smaller role, contributing at most 0.2 percentage points at $h = 3$.

5.3 Extended-sample evaluation

We next extend the analysis to the post-pandemic period 1990:Q1–2025:Q1. Due to exceptionally large forecast errors during the COVID-19 pandemic, we exclude calendar year 2020 from the out-of-sample loss computation, although the models are estimated on an expanding window including the pandemic period. Table A.3 reports the corresponding average negative log-scores for the AR-GARCH and SPF-GARCH specifications.

The findings broadly confirm the pre-pandemic results. SPF-based specifications are included in the MCS across all forecast horizons and weeks, demonstrating that the predictive content of the consensus forecast extends to the post-pandemic period. At the longest horizon $h = 3$ in week 4, however, all specifications including the baseline AR-GARCH enter the MCS, suggesting reduced power to discriminate between models when the pandemic period is included in the estimation window. While the NFCI adds some predictive improvements, the SPF-GARCH remains our preferred specification, especially at shorter horizons.

Figures B.10 and B.12 reveal the limits of the framework under extreme shocks. For GDP growth at the nowcast horizon, the SPF-GARCH model performs remarkably well, with the consensus forecast anticipating both the severity of the 2020 downturn and the timing of the subsequent rebound. At $h \geq 1$, however, the large pandemic forecast errors propagate through the conditional variance, producing predictive densities so wide that the scenario analysis becomes uninformative. Unemployment shows a similar pattern, with the nowcast remaining reasonably tight but longer-horizon densities widening dramatically.

By contrast, Figure B.11 shows that the scenario analysis remains informative for inflation throughout the pandemic period. At the nowcast horizon, the conditional and marginal densities track realized inflation closely. The scenario 90% quantile signaled elevated upside risk in real time, mostly exceeding the realized outcome during the 2021–2022 inflation surge. At $h \geq 1$, the scenario shifts the entire predictive density upward, but the upward shift remains plausible: even the three-step-ahead lower scenario quantile stayed close to the realized outcome when inflation declined in 2023. These findings likely reflect the relatively small forecast errors for inflation during the pandemic and the small correlations between inflation and the conditioning variables.

6 Conclusion

This paper proposes a framework for assessing joint macroeconomic tail risks by modeling the multivariate predictive density of real GDP growth, inflation, and unemployment. We combine skew-t marginals in an AR-GARCH framework with a Student's t-copula for the joint dependence, and SPF consensus forecasts as the key predictor in the conditional mean. Scenarios are based on conditional predictive distributions, for example the conditional quantiles of GDP growth given adverse outcomes in inflation and unemployment, and the joint predictive density itself is subject to formal out-of-sample evaluation.

Incorporating SPF median forecasts significantly improves joint density accuracy across horizons, with the largest gains realized once the mid-quarter SPF forecasts become

available. Conditional on the SPF, the marginals show distinct patterns at longer forecast horizons—GDP growth is fat-tailed but roughly symmetric, unemployment is fat-tailed and positively skewed with skewness strengthening over the horizon, and inflation is essentially symmetric—while at the nowcast horizon the predictive densities are close to Gaussian across all three indicators. Conditional scenario densities differ substantially from their marginal counterparts, with the strong negative correlation between GDP growth and unemployment as the main driver and tail dependence contributing persistently after the financial crisis. Together, these results underline the value of combining survey-based expectations with parametric density modeling for real-time macroeconomic risk assessment.

Overall, the framework offers policymakers a statistically grounded, real-time tool to assess joint macroeconomic risks across multiple dimensions. The approach remains deliberately agnostic: it does not deliver structural interpretation, but provides a transparent econometric basis for constructing scenario densities and for incorporating external forecasts or judgment through the conditional mean.

The COVID-19 pandemic produced unprecedented shocks to real activity that the framework struggled to accommodate, particularly for GDP growth and unemployment, while the predictive densities for inflation remained informative through the 2021–2022 surge. Extensions could accommodate regime shifts or richer dynamics in the dependence structure, or adopt outlier-robust estimation approaches. The joint system could also be broadened to include additional macro-financial indicators such as the term spread or credit conditions, either as standalone predictors or in combination with SPF projections where available.

References

Adams, P. A., T. Adrian, N. Boyarchenko, and D. Giannone (2021). Forecasting macroeconomic risks. *International Journal of Forecasting 37*(3), 1173–1191.

Adrian, T., N. Boyarchenko, and D. Giannone (2019). Vulnerable growth. *American Economic Review 109*(4), 1263–89.

Adrian, T. and M. K. Brunnermeier (2016). CoVaR. *The American Economic Review 106*(7), 1705.

Adrian, T., D. Giannone, M. Luciani, and M. West (2025). Scenario synthesis and macroeconomic risk. *arXiv preprint arXiv:2505.05193*.

Adrian, T., F. Grinberg, N. Liang, S. Malik, and J. Yu (2022). The term structure of growth-at-risk. *American Economic Journal: Macroeconomics 14*(3), 283–323.

Amburgey, A. J. and M. W. McCracken (2023). On the real-time predictive content of financial condition indices for growth. *Journal of Applied Econometrics 38*(2), 137–163.

Andrade, P., E. Ghysels, and J. Idier (2014). Inflation risk measures and their informational content. *Available at SSRN 2439607*.

Ang, A., G. Bekaert, and M. Wei (2007). Do macro variables, asset markets, or surveys forecast inflation better? *Journal of Monetary Economics 54*(4), 1163–1212.

Antolín-Díaz, J., I. Petrella, and J. F. Rubio-Ramírez (2021). Structural scenario analysis with SVARs. *Journal of Monetary Economics 117*, 798–815.

Azzalini, A. and A. Capitanio (2003). Distributions generated by perturbation of symmetry with emphasis on a multivariate skew t-distribution. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 367–389.

Bańbura, M., D. Giannone, and M. Lenza (2015). Conditional forecasts and scenario analysis with vector autoregressions for large cross-sections. *International Journal of Forecasting 31*(3), 739–756.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of econometrics 31*(3), 307–327.

Botelho, V., C. Foroni, and A. Renzetti (2024). Labour at risk. *European Economic Review 170*, 104849.

Brave, S. and R. A. Butters (2012). Diagnosing the financial system: Financial conditions and financial stress. *International Journal of Central Banking 8*(2), 191–239.

Brownlees, C. and A. B. Souza (2021). Backtesting global growth-at-risk. *Journal of Monetary Economics 118*, 312–330.

Carriero, A., T. E. Clark, and M. Marcellino (2022). Nowcasting tail risk to economic activity at a weekly frequency. *Journal of Applied Econometrics 37*(5), 843–866.

Carriero, A., T. E. Clark, and M. Marcellino (2024). Capturing macro-economic tail risks with bayesian vector autoregressions. *Journal of Money, Credit and Banking 56*(5), 1099–1127.

Chavleishvili, S., R. F. Engle, S. Fahr, M. Kremer, F. Lund-Thomsen, S. Manganelli, and B. Schwaab (2026). Macro-prudential policy under asymmetric risks: A bayesian structural quantile VAR approach. *Journal of Econometrics*. Forthcoming.

Chavleishvili, S. and S. Manganelli (2024). Forecasting and stress testing with quantile vector autoregression. *Journal of Applied Econometrics 39*(1), 66–85.

Clements, M. P., R. W. Rich, and J. S. Tracy (2023). Surveys of professionals. In *Handbook of Economic Expectations*, pp. 71–106. Academic Press.

Conrad, C. and O. Kleen (2020). Two are better than one: Volatility forecasting using multiplicative component garch-midas models. *Journal of Applied Econometrics 35*(1), 19–45.

Corradi, V. and J. Llorens-Terrazas (2026). Monitoring joint tail risks: An application to growth and inflation. *Journal of Econometrics 255*, 106237.

Delle Monache, D., A. De Polis, and I. Petrella (2024). Modeling and forecasting macroeconomic downside risk. *Journal of Business & Economic Statistics 42*(3), 1010–1025.

Diebold, F. X. and R. S. Mariano (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics 13*(3), 253–263.

Diks, C., V. Panchenko, O. Sokolinskiy, and D. van Dijk (2014). Comparing the accuracy of multivariate density forecasts in selected regions of the copula support. *Journal of Economic Dynamics and Control 48*, 79–94.

Diks, C., V. Panchenko, and D. van Dijk (2011). Likelihood-based scoring rules for comparing density forecasts in tails. *Journal of Econometrics 163*(2), 215–230.

Dimitriadis, T. and Y. Hoga (2025). Dynamic CoVaR modeling and estimation. *Journal of Business & Economic Statistics* (just-accepted).

Engle, R. (2002). Dynamic conditional correlation. *Journal of Business & Economic Statistics 20*(3), 339–350.

Engle, R. F., E. Ghysels, and B. Sohn (2013). Stock market volatility and macroeconomic fundamentals. *The Review of Economics and Statistics 95*(3), 776–797.

Faust, J. and J. H. Wright (2013). Forecasting inflation. In *Handbook of Economic Forecasting*, Volume 2, pp. 2–56. Elsevier.

Ferrara, L., M. Mogliani, and J.-G. Sahuc (2022). High-frequency monitoring of growth at risk. *International Journal of Forecasting 38*(2), 582–595.

Fissler, T. and Y. Hoga (2024). Backtesting systemic risk forecasts using multi-objective elicitability. *Journal of Business & Economic Statistics 42*(2), 485–498.

Ganics, G., B. Rossi, and T. Sekhposyan (2024). From fixed-event to fixed-horizon density forecasts: Obtaining measures of multihorizon uncertainty from survey density forecasts. *Journal of Money, Credit and Banking 56*(7), 1675–1704.

Giacomini, R. and H. White (2006). Tests of conditional predictive ability. *Econometrica 74*(6), 1545–1578.

Giglio, S., B. Kelly, and S. Pruitt (2016). Systemic risk and the macroeconomy: An empirical evaluation. *Journal of Financial Economics 119*(3), 457–471.

Girardi, G. and A. T. Ergün (2013). Systemic risk measurement: Multivariate GARCH estimation of CoVaR. *Journal of Banking & Finance 37*(8), 3169–3180.

Glas, A. (2020). Five dimensions of the uncertainty–disagreement linkage. *International Journal of Forecasting 36*(2), 607–627.

Gneiting, T. and A. E. Raftery (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association 102*(477), 359–378.

González-Rivera, G., J. Maldonado, and E. Ruiz (2019). Growth in stress. *International Journal of Forecasting 35*(3), 948–966. Forecasting issues in developing economies.

González-Rivera, G., C. V. Rodríguez-Caballero, and E. Ruiz (2024). Expecting the unexpected: Stressed scenarios for economic growth. *Journal of Applied Econometrics 39*(5), 926–942.

Hansen, B. E. (1994). Autoregressive conditional density estimation. *International Economic Review 35*(3), 705–730.

Hansen, P. R., A. Lunde, and J. M. Nason (2011). The model confidence set. *Econometrica 79*(2), 453–497.

Herbst, E., S. R. Konzem, and C. Scofield (2026). Alternative scenarios at the federal reserve from 1968 to 2020: Data, interpretation, and evaluation.

Krüger, F., T. E. Clark, and F. Ravazzolo (2017). Using entropic tilting to combine BVAR forecasts with external nowcasts. *Journal of Business & Economic Statistics 35*(3), 470–485.

López-Salido, D. and F. Loria (2024). Inflation at risk. *Journal of Monetary Economics 145*, 103570.

Rossi, B. and T. Sekhposyan (2019). Alternative tests for correct specification of conditional predictive densities. *Journal of Econometrics 208*(2), 638–657.

Schick, M. (2024). Nowcasting growth-at-risk using the Survey of Professional Forecasters. *Available at SSRN 4859937*.

Sheppard, K. (2009). MFE MATLAB function reference financial econometrics.

Stark, T. (2010). Realistic evaluation of real-time forecasts in the Survey of Professional Forecasters. *Research Rap Special Report* (May).

Waggoner, D. F. and T. Zha (1999). Conditional forecasts in dynamic multivariate models. *The Review of Economics and Statistics 81*(4), 639–651.

Appendices

A Tables

Table A.1: Unconditional sample correlations

	Panel (a): 1990 – 2019		
	Real GDP growth	Inflation	Unemployment
Real GDP growth	1.00		
Inflation	0.03	1.00	
Unemployment rate	-0.17	-0.14	1.00

	Panel (b): 1990 – 2024		
	Real GDP growth	Inflation	Unemployment
Real GDP growth	1.00		
Inflation	0.22	1.00	
Unemployment rate	-0.21	-0.28	1.00

Notes: Unconditional sample correlations between the three indicators. Panels and sample as in Table 2.

Table A.2: In-sample log-likelihoods for alternative distributional specifications

		$h = 3$		$h = 2$	
Copula	Marginal	$w = 4$	$w = 8$	$w = 4$	$w = 8$
Normal	Normal	-946.51	-943.08	-893.05	-856.90
t	t	-941.25**	-929.36***	-886.91**	-853.30
t	Skew-t	-909.77***	-912.33***	-865.71***	-841.13***
		$h = 1$		$h = 0$	
Copula	Marginal	$w = 4$	$w = 8$	$w = 4$	$w = 8$
Normal	Normal	-827.09	-777.29	-609.41	-532.47
t	t	-821.10**	-770.48***	-603.83**	-529.45
t	Skew-t	-807.41***	-764.23***	-599.25***	-518.62***

Notes: This table reports in-sample log-likelihoods of the SPF-GARCH model under three nested distributional specifications, estimated on the full sample 1968:Q4–2019:Q4. The baseline is a Gaussian copula with normal marginals (Normal/Normal). The second specification uses a Student's t-copula with symmetric Student's t marginals (t/t), introducing four additional parameters. The third specification uses skew-t marginals together with the Student's t-copula (t/Skew-t), introducing seven additional parameters. Stars indicate the significance of likelihood-ratio tests against the Normal/Normal baseline, with four and seven restrictions for the second and third rows, respectively. Columns indicate the forecast horizon h (in quarters) and the week w of the quarter in which forecasts are formed. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

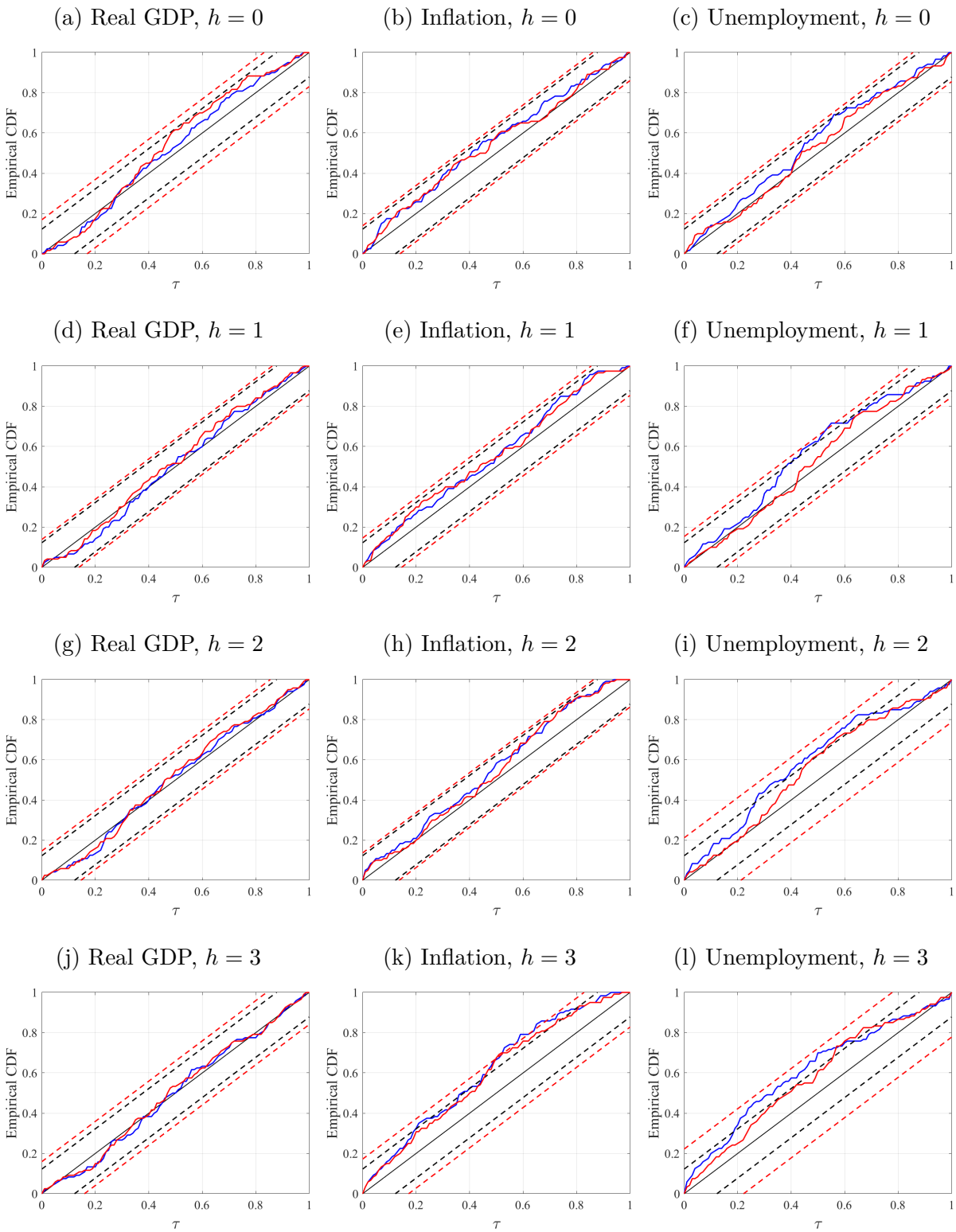
Table A.3: Out-of-sample log-scores excluding the COVID-19 pandemic

	$h = 3$		$h = 2$		$h = 1$		$h = 0$	
week	4	8	4	8	4	8	4	8
AR-GARCH:	5.70	5.66	5.24	4.88	4.49	4.34	3.66	3.09
$+x^{\mu}_{t,w,i}$ = NFCI	**5.03**	5.04	4.81	4.78	4.29	4.20	3.50	3.04
$+x^{\sigma}_{t,w,i}$ = NFCI	5.75	5.58	5.14	4.79	4.48	4.40	3.68	3.12
SPF-GARCH:	5.39	4.86	4.83	4.51	4.23	**3.84**	3.51	2.67
$+x^{\mu}_{t,w,i}$ = NFCI	5.09	4.90	4.72	4.52	**4.19**	3.91	**3.42**	**2.67**
$+x^{\sigma}_{t,w,i}$ = NFCI	5.31	**4.83**	**4.69**	**4.40**	4.22	4.04	3.50	2.73
$+x^{\sigma}_{t,w,i}$ = SPF^{IQR}	5.40	4.90	4.91	4.52	4.23	3.87	3.55	2.74

Notes: This table reports average negative log-scores of the joint predictive density of GDP growth, inflation, and unemployment over the out-of-sample period 1990:Q1–2025:Q1, excluding the 2020 calendar year (137 observations). All specifications use skew-t marginals and a Student's t-copula. AR-GARCH denotes the baseline AR(1)–GARCH(1,1) model; SPF-GARCH augments the conditional mean with the SPF consensus forecast. The NFCI and the SPF interquartile range (IQR) are the real-time series constructed in Section 2. All models are estimated using an expanding window with an initial length of 85 observations. Columns indicate the forecast horizon h (in quarters) and the week w of the quarter in which forecasts are formed. Gray shading denotes inclusion in the 90% model confidence set; bold entries indicate the lowest average loss within each column.

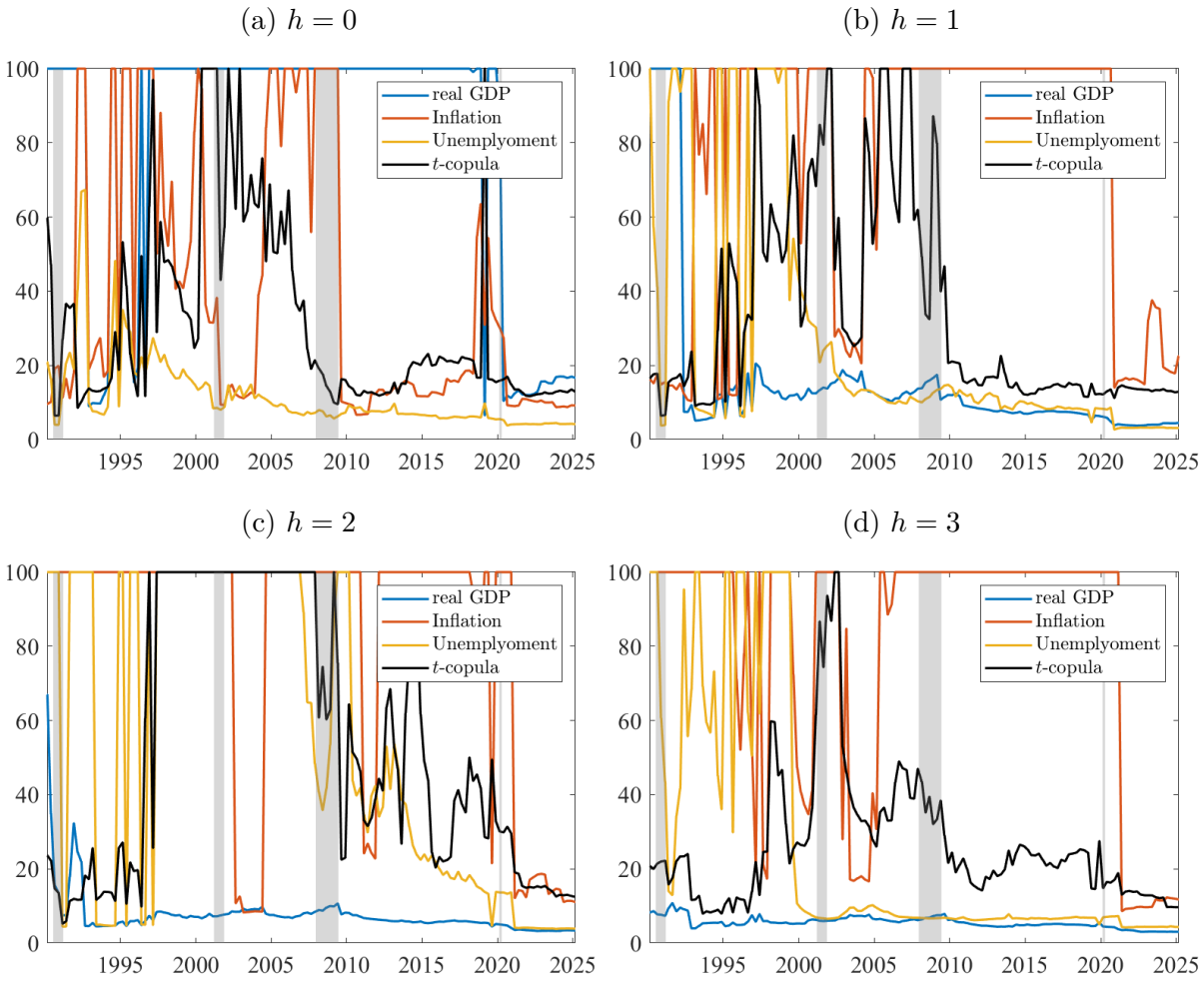
B Figures

Figure B.1: Probability integral transform of the SPF-GARCH model



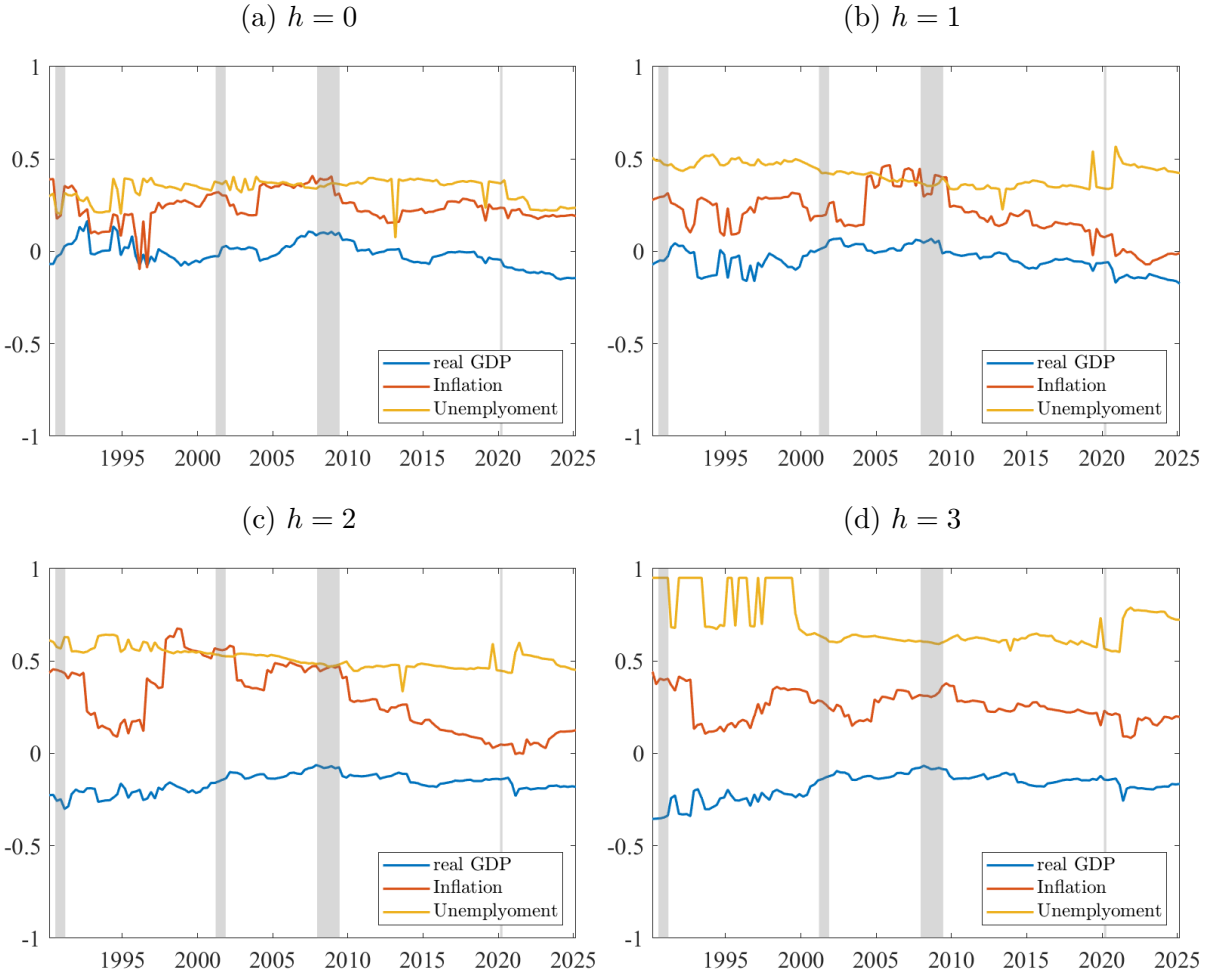
Notes: This figure displays the probability integral transform (PIT) of the SPF-GARCH predictive density over 1990:Q1–2019:Q4, estimated on an expanding window with skew-t marginals and a Student's t-copula. Solid blue lines show PITs for forecasts formed in week 4, and solid red lines for week 8. Black dashed lines denote 95% confidence bands under the null of uniformity and independence. Red dashed lines show 95% bootstrap bands assuming uniformity only, following Rossi and Sekhposyan (2019).

Figure B.2: Estimated degrees of freedom from the SPF-GARCH model



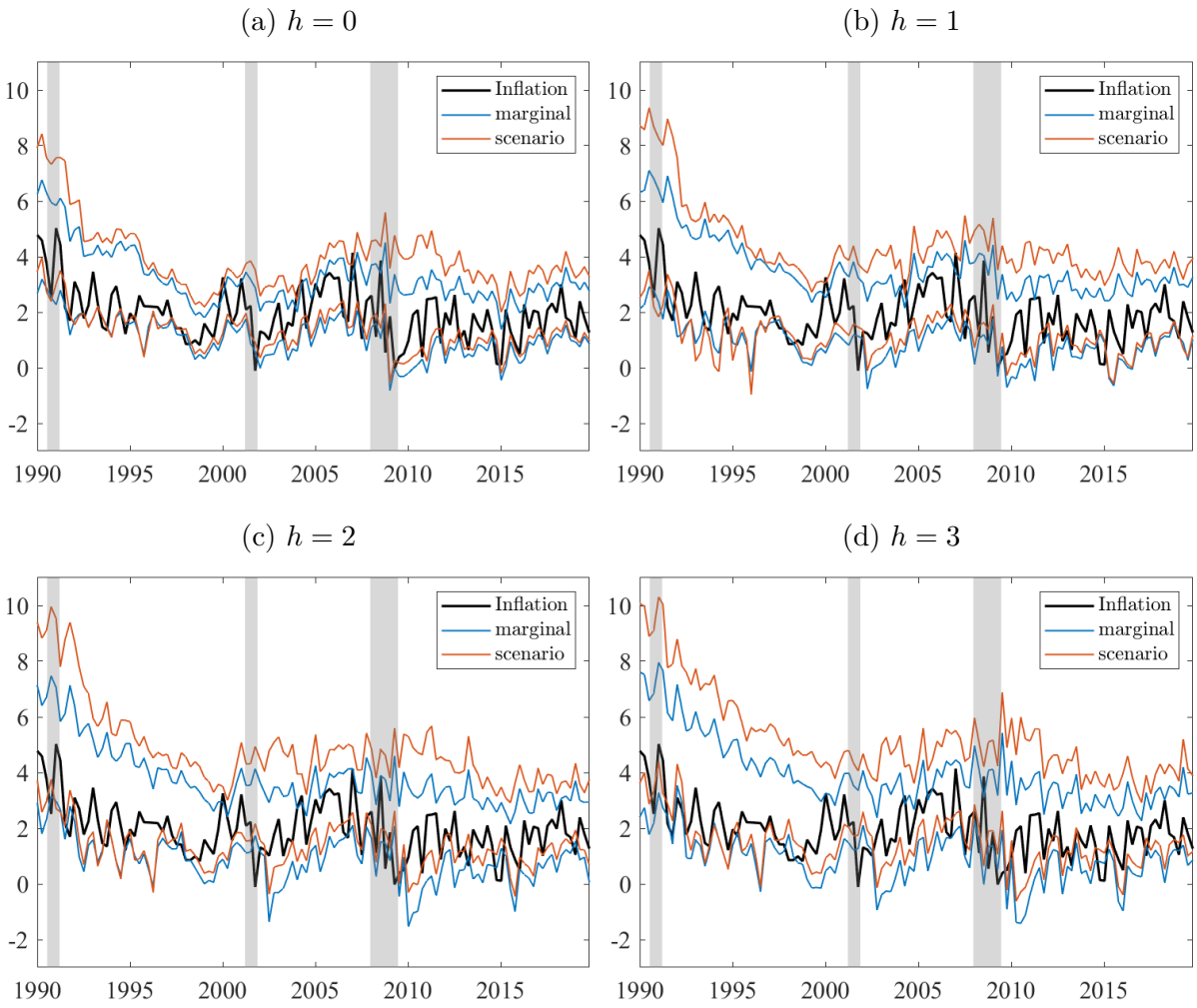
Notes: This figure shows the estimated degrees-of-freedom parameters of the skew-t marginals and the Student's t-copula in the SPF-GARCH model. Blue, red, and yellow lines correspond to the degrees-of-freedom parameters of the marginal distributions for real GDP growth, inflation, and unemployment, respectively. The black line shows the degrees-of-freedom parameter of the Student's t-copula. The sample period is 1990:Q1–2025:Q1 and the model is estimated using an expanding window with an initial length of 85 observations, evaluated in week 8 of each quarter. Vertical gray bands indicate NBER recessions.

Figure B.3: Estimated skewness parameters from the SPF-GARCH model



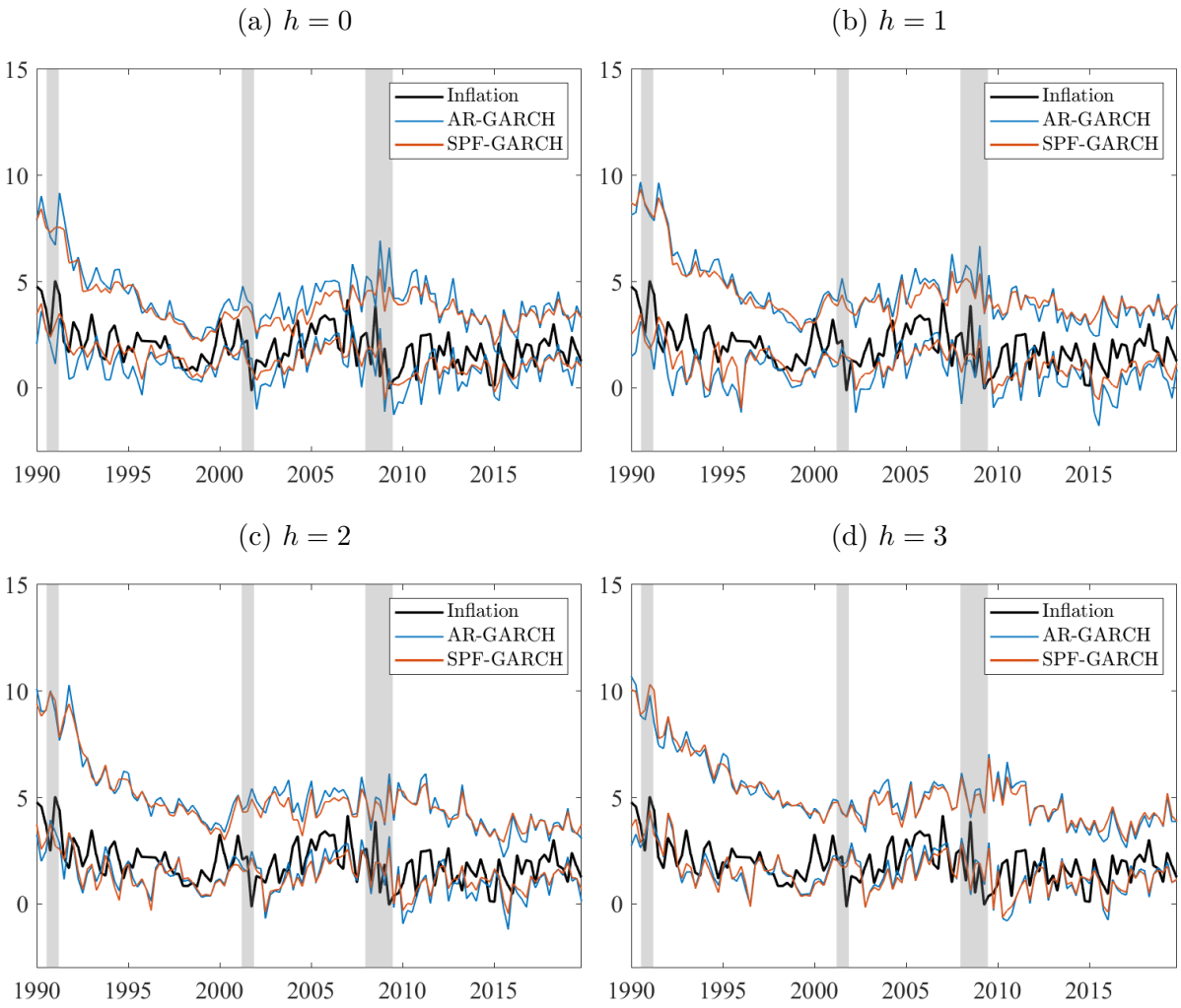
Notes: This figure shows the estimated skewness parameters of the skew-t marginals in the SPF-GARCH model. Blue, red, and yellow lines correspond to the skewness parameters of the marginal distributions for real GDP growth, inflation, and unemployment, respectively. The sample period is 1990:Q1–2025:Q1 and the model is estimated using an expanding window with an initial length of 85 observations, evaluated in week 8 of each quarter. Vertical gray bands indicate NBER recessions.

Figure B.4: Upside inflation scenarios based on the SPF-GARCH model



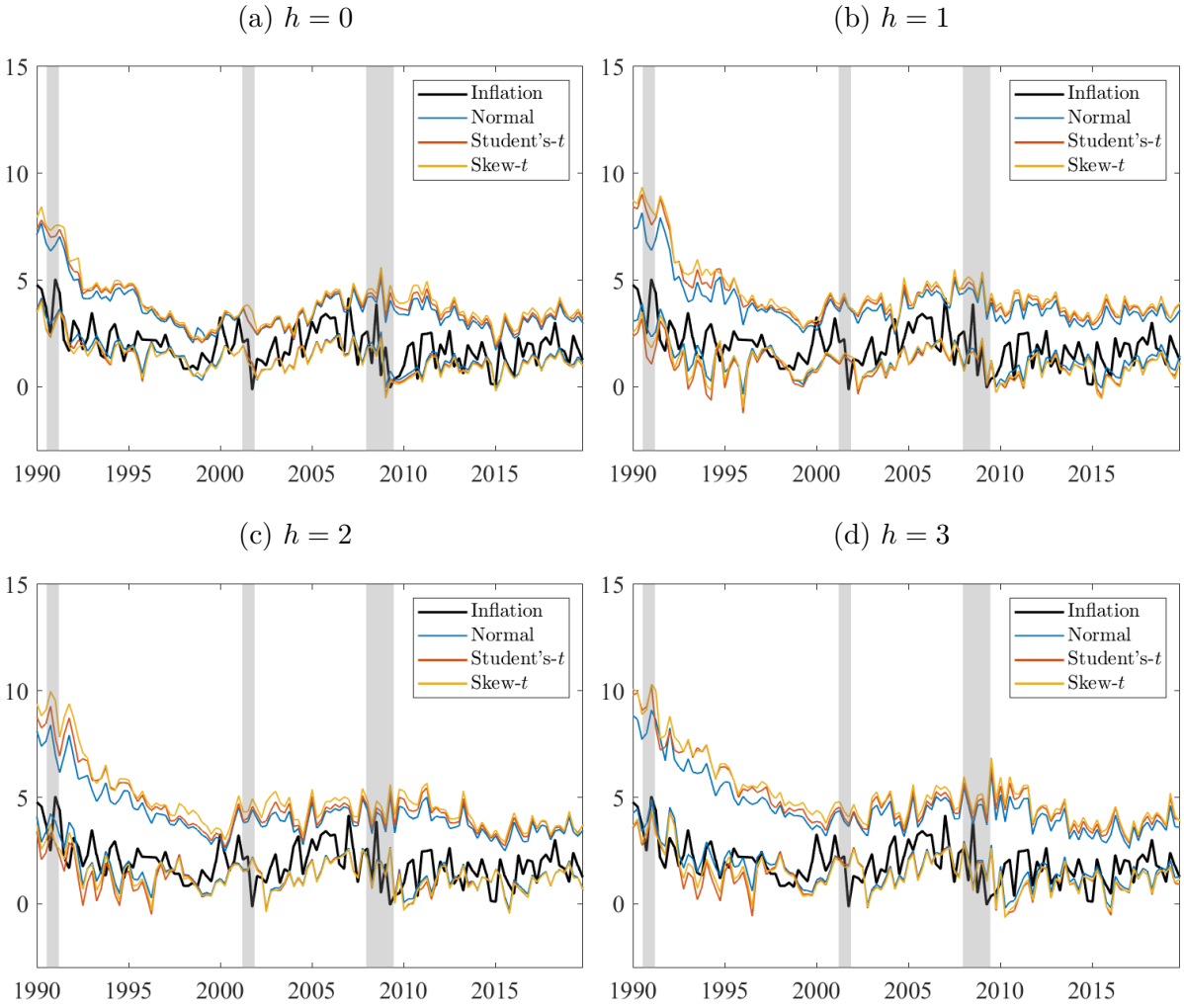
Notes: This figure plots the 10th and 90th percentiles of the marginal predictive density for inflation (blue lines) and the corresponding conditional percentiles when both GDP growth and unemployment are in their lower deciles (red lines), i.e., $Q^{\pi}_{t+h,w}(\tau \,|\, y \leq Q^{y}_{t+h,w}(0.1),\, u \leq Q^{u}_{t+h,w}(0.1))$ for $\tau \in \{0.1, 0.9\}$. The black line depicts realized quarterly annualized inflation. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. All forecasts are from the SPF-GARCH model with skew-t marginals and a Student's t-copula. The sample period is 1990:Q1–2019:Q4, and the model is estimated using an expanding window with an initial length of 85 observations. Vertical gray bands indicate NBER recessions.

Figure B.5: Upside inflation scenarios under AR-GARCH and SPF-GARCH models



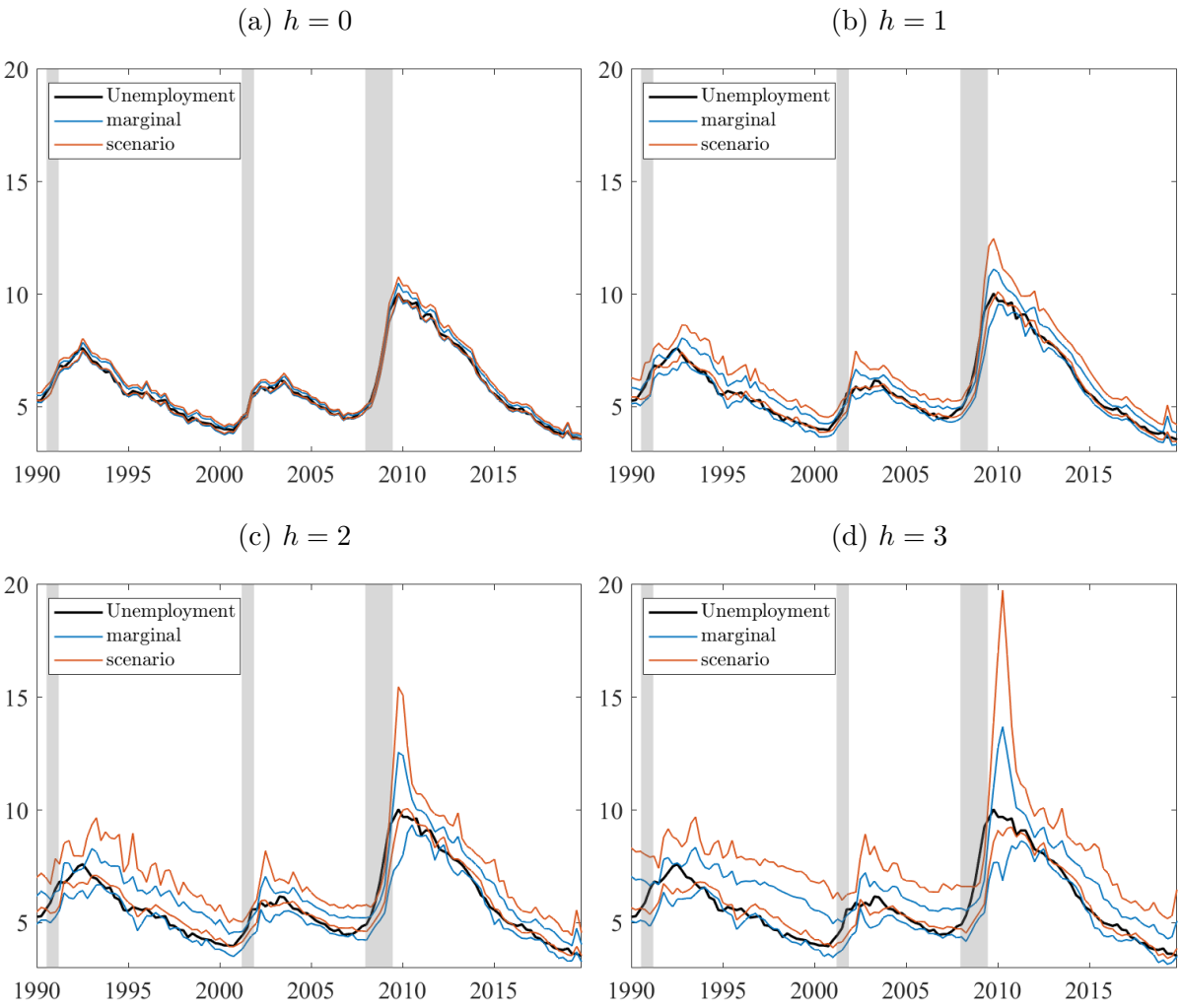
Notes: This figure plots the 10th and 90th percentiles of the marginal predictive density for inflation (blue lines) and the corresponding conditional percentiles when both GDP growth and unemployment are in their lower deciles (red lines), i.e., $Q^{\pi}_{t+h,w}(\tau \,|\, y \leq Q^{y}_{t+h,w}(0.1), u \leq Q^{u}_{t+h,w}(0.1))$ for $\tau \in \{0.1, 0.9\}$. The black line depicts realized quarterly annualized inflation. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. All forecasts are from the SPF-GARCH model with skew-t marginals and a Student's t-copula. The sample period is 1990:Q1–2019:Q4, and the model is estimated using an expanding window with an initial length of 85 observations. Vertical gray bands indicate NBER recessions.

Figure B.6: Upside inflation scenarios across distributional specifications



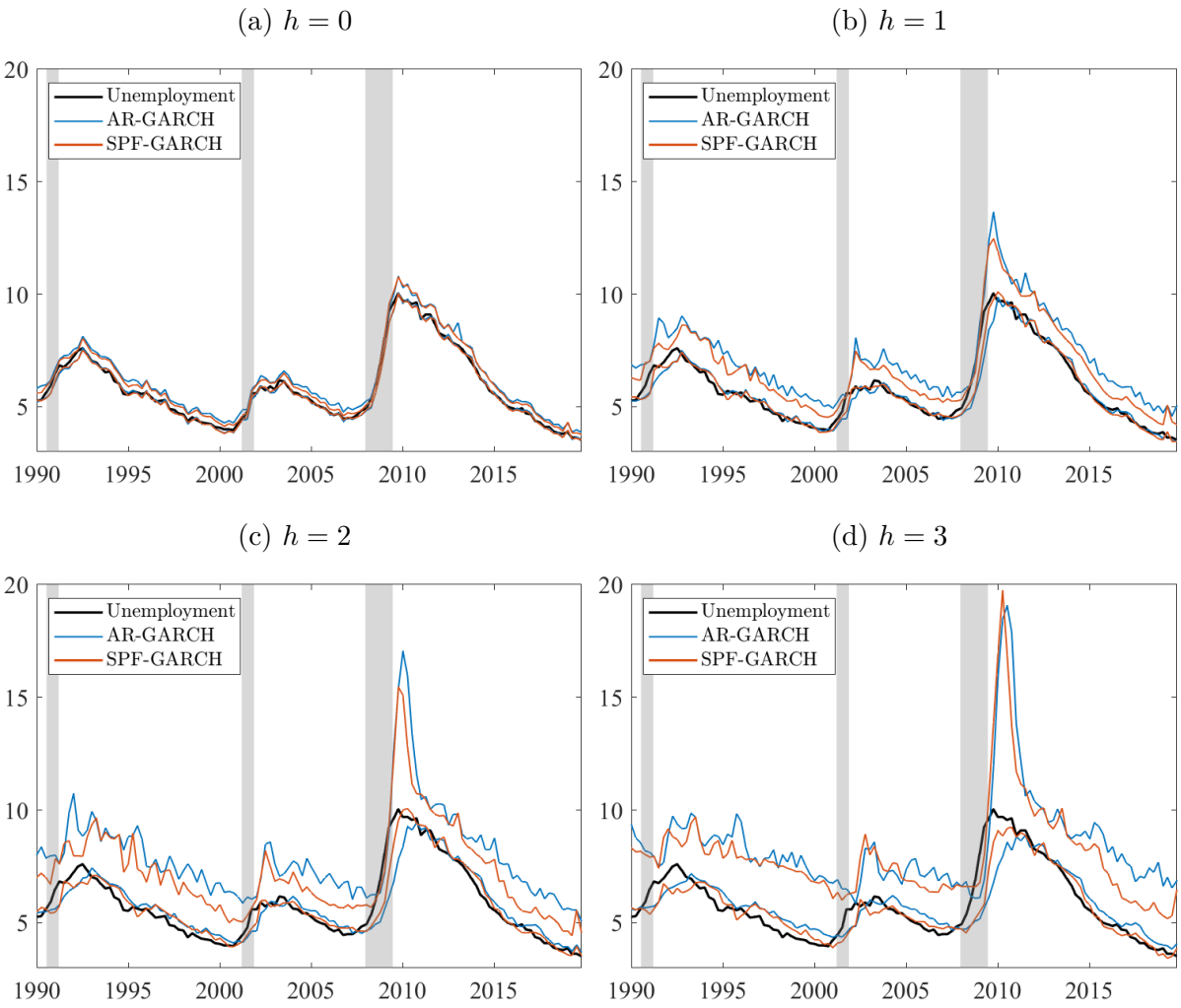
Notes: This figure plots the 10th and 90th percentiles of the conditional predictive density for inflation when both GDP growth and unemployment are in their lower deciles, i.e., $Q^{\pi}_{t+h,w}(\tau \mid y \leq Q^{y}_{t+h,w}(0.1), u \leq Q^{u}_{t+h,w}(0.1))$ for $\tau \in \{0.1, 0.9\}$, across three distributional specifications of the SPF-GARCH model: a Gaussian copula with normal marginals (blue), a Student's t-copula with symmetric Student's t marginals (red), and a Student's t-copula with skew-t marginals (yellow). The black line depicts realized quarterly annualized inflation. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. The sample period is 1990:Q1–2019:Q4, and the models are estimated using an expanding window with an initial length of 85 observations. Vertical gray bands indicate NBER recessions.

Figure B.7: Upside unemployment scenarios based on the SPF-GARCH model



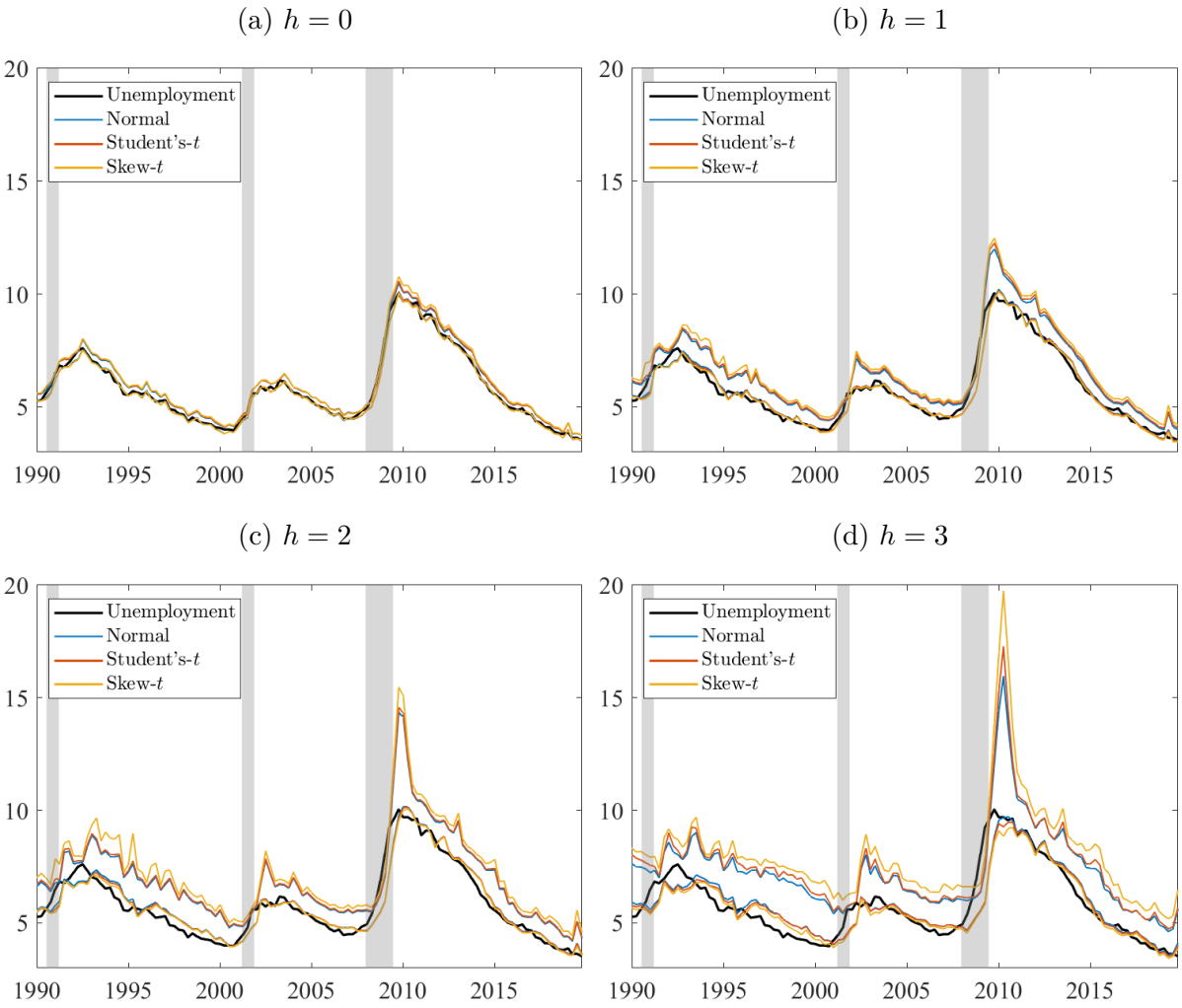
Notes: This figure plots the 10th and 90th percentiles of the marginal predictive density for unemployment (blue lines) and the corresponding conditional percentiles when both GDP growth and inflation are in their lower deciles (red lines), i.e., $Q^u_{t+h,w}(\tau \,|\, y \leq Q^y_{t+h,w}(0.1),\, \pi \leq Q^\pi_{t+h,w}(0.1))$ for $\tau \in \{0.1, 0.9\}$. The black line depicts realized quarterly unemployment. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. All forecasts are from the SPF-GARCH model with skew-t marginals and a Student's t-copula. The sample period is 1990:Q1–2019:Q4, and the model is estimated using an expanding window with an initial length of 85 observations. Vertical gray bands indicate NBER recessions.

Figure B.8: Upside unemployment scenarios under AR-GARCH and SPF-GARCH models



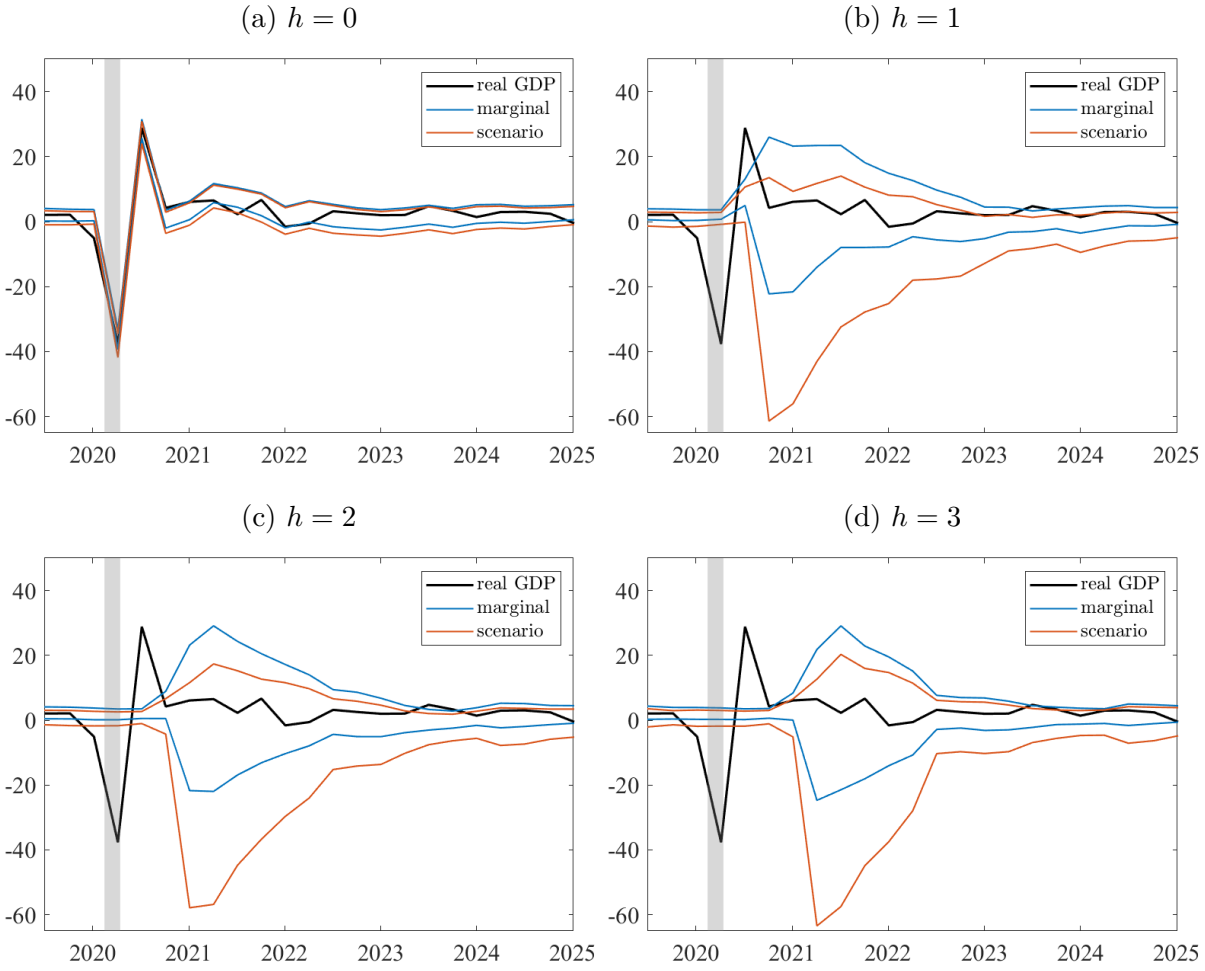
Notes: This figure plots the 10th and 90th percentiles of the marginal predictive density for unemployment (blue lines) and the corresponding conditional percentiles when both GDP growth and inflation are in their lower deciles (red lines), i.e., $Q^u_{t+h,w}(\tau \,|\, y \leq Q^y_{t+h,w}(0.1),\, \pi \leq Q^\pi_{t+h,w}(0.1))$ for $\tau \in \{0.1, 0.9\}$. The black line depicts realized quarterly unemployment. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. All forecasts are from the SPF-GARCH model with skew-t marginals and a Student's t-copula. The sample period is 1990:Q1–2019:Q4, and the model is estimated using an expanding window with an initial length of 85 observations. Vertical gray bands indicate NBER recessions.

Figure B.9: Upside unemployment scenarios across distributional specifications



Notes: This figure plots the 10th and 90th percentiles of the conditional predictive density for unemployment when both GDP growth and inflation are in their lower deciles, i.e., $Q^u_{t+h,w}(\tau \,|\, y \leq Q^y_{t+h,w}(0.1), \pi \leq Q^\pi_{t+h,w}(0.1))$ for $\tau \in \{0.1, 0.9\}$, across three distributional specifications of the SPF-GARCH model: a Gaussian copula with normal marginals (blue), a Student's t-copula with symmetric Student's t marginals (red), and a Student's t-copula with skew-t marginals (yellow). The black line depicts realized quarterly unemployment. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. The sample period is 1990:Q1–2019:Q4, and the models are estimated using an expanding window with an initial length of 85 observations. Vertical gray bands indicate NBER recessions.

Figure B.10: Downside GDP scenarios during the COVID-19 period



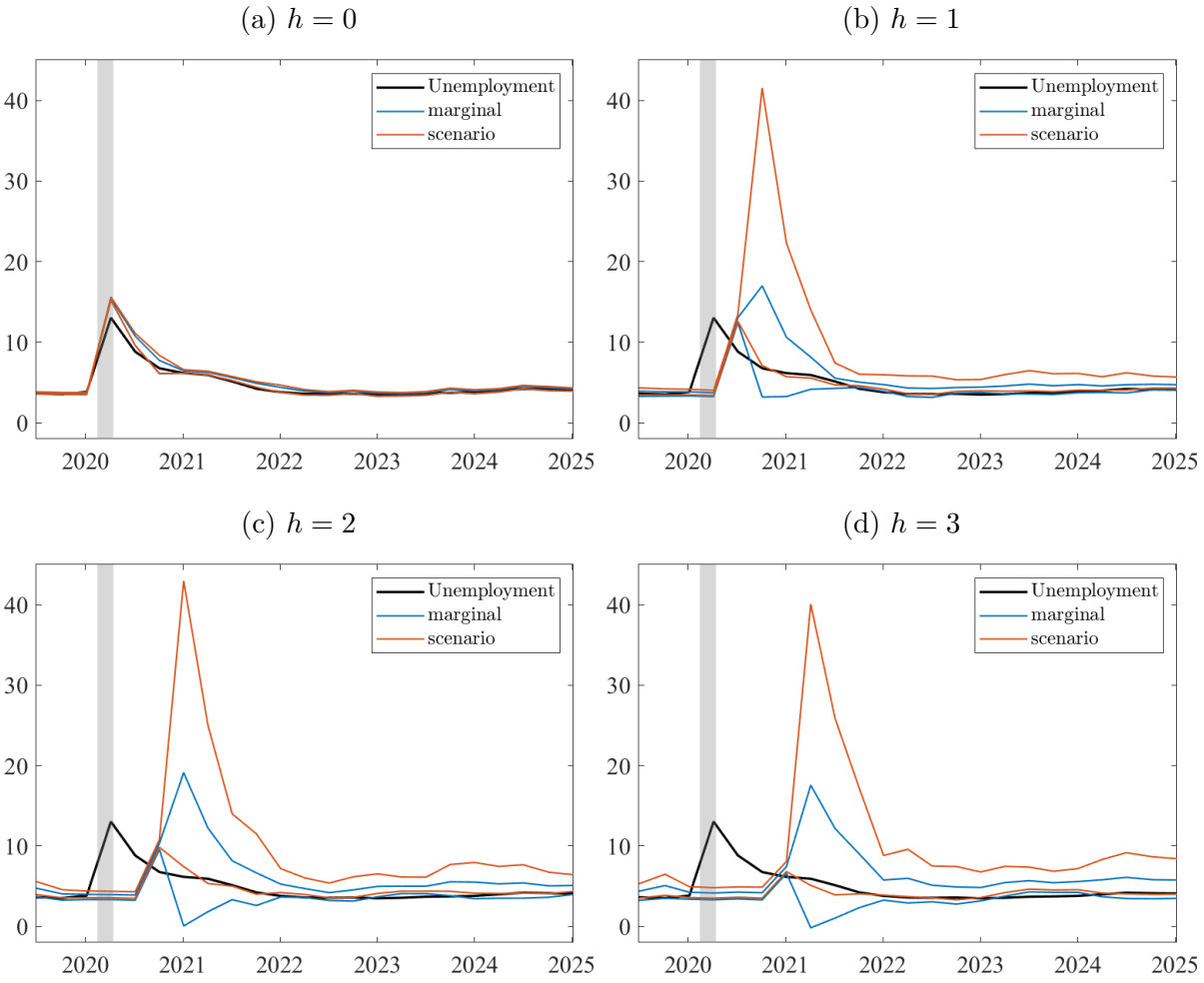
Notes: This figure plots the 10th and 90th percentiles of the marginal predictive density for real GDP growth (blue lines) and the corresponding conditional percentiles when both inflation and unemployment are in their upper deciles (red lines), i.e., $Q^{y}_{t+h,w}(\tau \,|\, \pi \geq Q^{\pi}_{t+h,w}(0.9),\, u \geq Q^{u}_{t+h,w}(0.9))$ for $\tau \in \{0.1, 0.9\}$. The black line depicts realized quarterly annualized GDP growth. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. All forecasts are from the SPF-GARCH model with skew-t marginals and a Student's t-copula, estimated using an expanding window. The sample period shown is 2019:Q3–2025:Q1. Vertical gray bands indicate NBER recessions.

Figure B.11: Upside inflation scenarios during the COVID-19 period



Notes: This figure plots the 10th and 90th percentiles of the marginal predictive density for inflation (blue lines) and the corresponding conditional percentiles when both GDP growth and unemployment are in their lower deciles (red lines), i.e., $Q^{\pi}_{t+h,w}(\tau \,|\, y \leq Q^{y}_{t+h,w}(0.1),\, u \leq Q^{u}_{t+h,w}(0.1))$ for $\tau \in \{0.1, 0.9\}$. The black line depicts realized quarterly annualized inflation. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. All forecasts are from the SPF-GARCH model with skew-t marginals and a Student's t-copula, estimated using an expanding window. The sample period shown is 2019:Q3–2025:Q1. Vertical gray bands indicate NBER recessions.

Figure B.12: Upside unemployment scenarios during the COVID-19 period



Notes: This figure plots the 10th and 90th percentiles of the marginal predictive density for unemployment (blue lines) and the corresponding conditional percentiles when both GDP growth and inflation are in their lower deciles (red lines), i.e., $Q^u_{t+h,w}(\tau \,|\, y \leq Q^y_{t+h,w}(0.1), \pi \leq Q^\pi_{t+h,w}(0.1))$ for $\tau \in \{0.1, 0.9\}$. The black line depicts realized quarterly unemployment. Each panel corresponds to a different forecast horizon h, with forecasts evaluated in week $w = 8$ of each quarter. All forecasts are from the SPF-GARCH model with skew-t marginals and a Student's t-copula, estimated using an expanding window. The sample period shown is 2019:Q3–2025:Q1. Vertical gray bands indicate NBER recessions.

C Hansen's skew-t distribution

We use the skew-t distribution of Hansen (1994), reparameterized to have mean zero and unit variance. The distribution is characterized by two parameters: $\nu > 2$ governs the tail thickness, and $\lambda \in (-1, 1)$ controls the skewness. It reduces to a standardized Student's t distribution when $\lambda = 0$, and to the standard normal as $\lambda \to 0$ and $\nu \to \infty$.

Define the constants

$$c = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi(\nu-2)}\,\Gamma\left(\frac{\nu}{2}\right)}, \qquad a = 4\lambda c\,\frac{\nu-2}{\nu-1}, \qquad b = \sqrt{1+3\lambda^2-a^2}, \tag{26}$$

and the threshold $x^\star = -a/b$. The probability density function is then

$$f_{\mathrm{ST}}(x;\nu,\lambda) = \begin{cases} b\,c\left[1+\dfrac{1}{\nu-2}\left(\dfrac{bx+a}{1-\lambda}\right)^2\right]^{-\frac{\nu+1}{2}} & \text{if } x < x^\star, \\ b\,c\left[1+\dfrac{1}{\nu-2}\left(\dfrac{bx+a}{1+\lambda}\right)^2\right]^{-\frac{\nu+1}{2}} & \text{if } x \geq x^\star. \end{cases} \tag{27}$$

The parameters a and b ensure zero mean and unit variance, so that c acts as a Student's t–type normalizing constant.

The corresponding cumulative distribution function admits a semi-closed form in terms of the standardized Student's t CDF, denoted $t_\nu(\cdot)$:

$$F_{\mathrm{ST}}(x;\nu,\lambda) = \begin{cases} (1-\lambda)\,t_\nu\left(\sqrt{\frac{\nu}{\nu-2}}\,\dfrac{bx+a}{1-\lambda}\right) & \text{if } x < x^\star, \\ \dfrac{1-\lambda}{2} + (1+\lambda)\left[t_\nu\left(\sqrt{\frac{\nu}{\nu-2}}\,\dfrac{bx+a}{1+\lambda}\right) - \frac{1}{2}\right] & \text{if } x \geq x^\star. \end{cases} \tag{28}$$

The inverse CDF is obtained by inverting each branch of F_{ST} using the standardized Student's t inverse CDF. Sampling is straightforward via the probability integral transform: draw $u \sim U(0,1)$, then set $x = F_{\mathrm{ST}}^{-1}(u;\nu,\lambda)$.